12. Installation Instruction

12.1 Mono Bloc Unit

Attached Accessories

No.	Accessories part	Qty.
1	Drain elbow 	1
2	Rubber cap 	8
3	Remote controller 	1

Optional Accessory

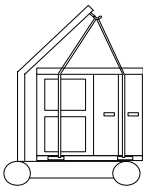
No.	Accessories part	Qty.
4	Optional PCB (CZ-NS4P)	1
5	Base Pan Heater (CZ-NE3P)	1
6	Network Adaptor (CZ-TAW1)	1
7	Network Adaptor Cable (CZ-TAW1-CBL)	1

Field Supply Accessories (Optional)

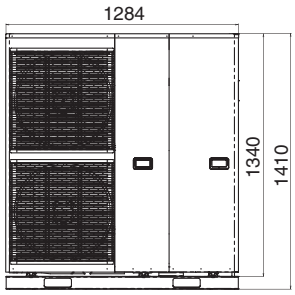
No.	Part		Model	Specification	Maker
i	2-way valve kit	Electromotoric Actuator	SFA21/18	AC230V	Siemens
		2-port Valve	VVI46/25	-	Siemens
ii	3-way valve kit	Electromotoric Actuator	SFA21/18	AC230V	Siemens
		3-port Valve	VXI46/25	-	Siemens
iii	Room thermostat	Wired	PAW-A2W-RTWIRED	AC230V	-
		Wireless	PAW-A2W-RTWIRELESS		
iv	Mixing valve	-	167032	AC230V	Caleffi
v	Pump	-	Yonos 25/6	AC230V	Wilo
vi	Buffer tank sensor	-	PAW-A2W-TSBU	-	-
vii	Outdoor sensor	-	PAW-A2W-TSOD	-	-
viii	Zone water sensor	-	PAW-A2W-TSHC	-	-
ix	Zone room sensor	-	PAW-A2W-TSRT	-	-
x	Solar sensor	-	PAW-A2W-TSSO	-	-

■ It is recommended to purchase the field supply accessories listed in above table.

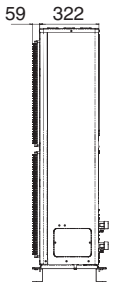
- Handling of Mono bloc unit
  - Mono bloc unit is a large and heavy apparatus. The handling of the unit only to be done by lifting tools with slings. These slings can be fitted into sleeves at the unit's base frame.



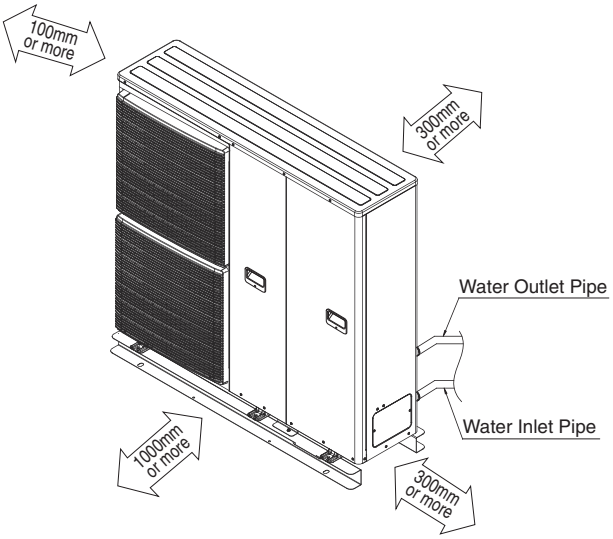
Dimension Diagram

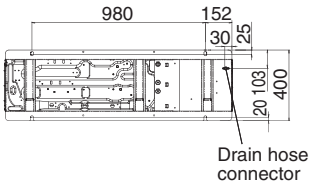


Front view

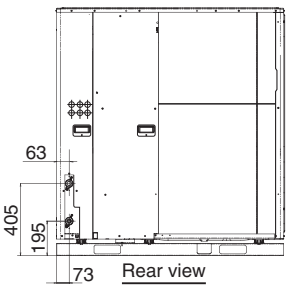


Side view





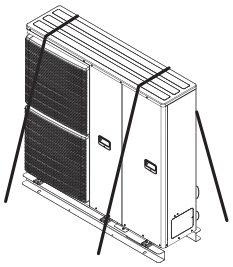
Bottom view



Rear view

12.1.1 Select the Best Location

- Install the Mono bloc unit in outdoor locations only.
- Avoid installations in areas where the ambient temperature may drop below -20°C.
- The Mono bloc unit must be installed on a flat, solid surface.
- A place removed from any heat source or steam which may effect the operation of the Mono bloc unit.
- A place where air circulation is good.
- A place where drainage can be easily done.
- A place where Mono bloc unit's operation noise will not cause discomfort to the user.
- A place which is accessible for maintenance.
- Ensure to keep minimum distance of spaces as illustrated above from wall, ceiling, or other obstacles.
- A place where flammable gas leaking might not occur.
- A place where the Mono bloc unit's piping and wiring lengths come within reasonable ranges.
- If an awning is built over the unit to prevent direct sunlight or rain, be careful that heat radiation from the condenser is not obstructed.
- Do not place any obstacles which may cause a short circuit of the discharged air.
- Avoid installing the Mono bloc unit at a location where suction side may be exposed directly to wind.
- If Mono bloc unit installed near sea, region with high content of sulphur or oily location (e.g. machinery oil, etc.), it lifespan maybe shorten.
- When installing the product in a place where it will be affected by typhoon or strong wind such as wind blowing between buildings, including the rooftop of a building and a place where there is no building in surroundings, fix the product with an overturn prevention wire, etc. (Overturn prevention fitting model number: K-KYZP15C)



12.1.2 Mono Bloc Unit Installation

Mono bloc unit will become heavy when filled with water. Please install the unit on a strong concrete floor and consider the weight of the unit and water.

- Fix Mono bloc unit on the concrete floor with M12 anchor bolt at 4 locations.
- Pull-out strength of these anchor bolts must be above 15000N.

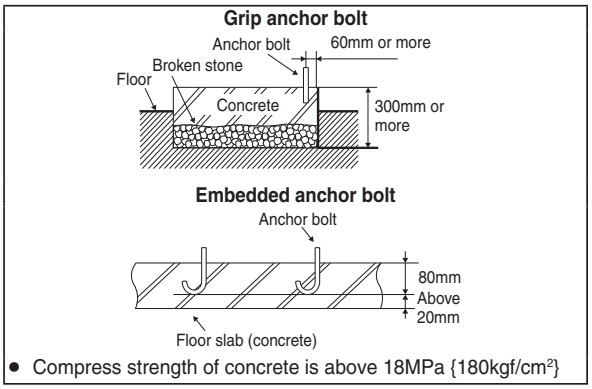
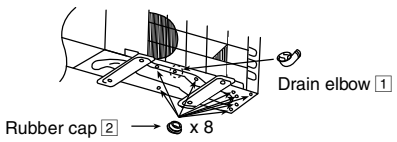


Illustration of grip type and embedded type anchor bolt

12.1.2.1 Disposal of Mono Bloc Unit Drain Water

- When a Drain elbow is used, please ensure to follow below:
  - The unit should be placed on a stand which is taller than 50mm.
  - Cover the 8 holes (ø20mm) with Rubber cap (refer to illustration below)
  - Use a tray (field supply) when necessary to dispose the Mono bloc unit drain water.

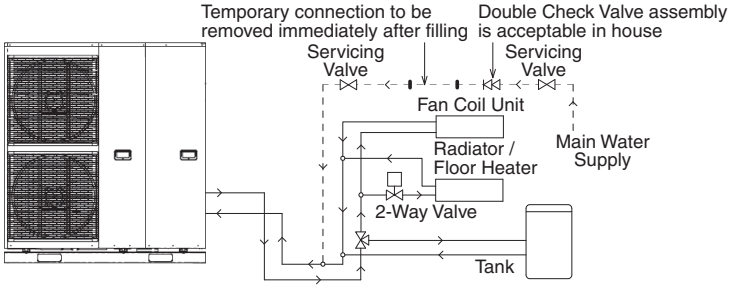


- If the unit is used in an area where temperature falls below 0°C for 2 or 3 consecutive days, it is recommended not to use the Drain elbow and Rubber cap, for the drain water freezes and the fan will not rotate.

12.1.3 Piping Installation

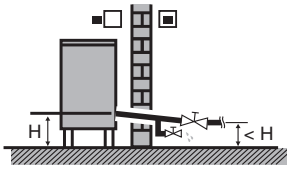
⚠ WARNING
This section is for authorized and licensed electrician / water system installer only. Work behind the Cabinet front plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

12.1.3.1 Typical Piping Installation



Please engage a licensed water circuit installer to install this water circuit.

- This water circuit must comply with relevant European and national regulations (including EN61770), and local building regulation codes.
- Ensure the components installed in the water circuit could withstand water pressure during operation.
- Do not apply excessive force to piping that may damage the pipes.
- Use Rp 1¼" nut for both water inlet and water outlet connection and clean all piping with tap water before connecting to the Mono bloc unit.
- Cover the pipe end to prevent dirt and dust when inserting it through a wall. If an existing tank is to be connected to this Mono bloc unit, ensure the pipes are clean before water pipe installation is carried out.
- Choose proper sealer which can withstand the pressures and temperatures of the system.
- Make sure to use two spanners to tighten the connection. Tighten the nuts with torque wrench: 117.6 N•m.
- If non-brass metallic piping is used for installation, make sure to insulate the piping to prevent galvanic corrosion.
- Do not use pipes that are crushed or deformed. If these inferior pipes are used, it may cause unit malfunction.
- Make sure to insulate the water circuit piping (insulator thickness : 20mm or more) to prevent condensation during cooling operation and reduction of heating capacity, as well as avoid freezing of the outdoor water circuit piping during winter season.
- After installation, check the water leakage condition in connection area during test run.
- In case of a power supply failure or pump operating failure, drain the system (as suggested in the figure below).



When water is idle inside the system, freezing up is very likely to happen which could damage the system.

Drainage piping installation

- Use a drain hose with inner diameter of 15 mm.
- The hose must be installed in a continuously downward direction and left open to the frost-free atmosphere.
- If drain hose is long, use a metal support fixture along the way to eliminate the wavy pattern of drain tube.
- Water will drip from this hose, therefore the outlet of this hose must be installed in an area where the outlet cannot become blocked.
- Do not insert this hose into sewage or drain pipe that may generate ammonia gas, sulfuric gas, etc.
- If necessary, use a hose clamp to tighten the hose at drain hose connector to prevent it from leaking.

⚠ CAUTION
Do not over tighten, over tightening cause water leakage.

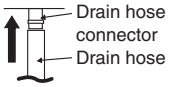


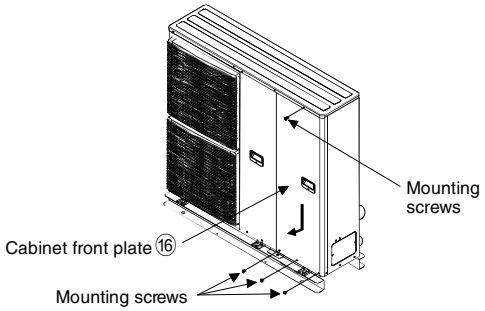
Illustration of how to fix drain hose to Mono bloc unit

12.1.4 Connect the Cable to Mono Bloc Unit

⚠ WARNING
This section is for authorised and licensed electrician only. Work behind the cabinet front plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

12.1.4.1 Remove The Cabinet Front Plate

1. Remove the 4 mounting screws as shown in the illustration.
2. Slide the cabinet front plate downward to release the pawls. Then, pull it toward front to remove it.



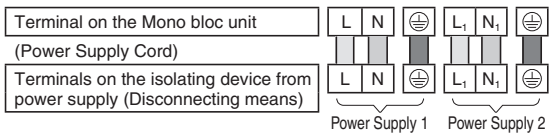
12.1.4.2 Fixing of Power Supply Cord

(REFER TO WIRING DIAGRAM AT UNIT FOR DETAIL)

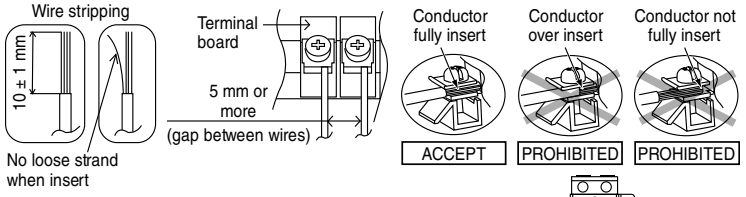
1. An isolating device must be connected to the power supply cable.
   - Isolating device (Disconnecting means) should have minimum 3.0 mm contact gap.
   - Connect the approved polychloroprene sheathed power supply 1 cord and power supply 2 cord and type designation 60245 IEC 57 or heavier cord to the terminal board, and to the other end of the cord to isolating device (Disconnecting means). See below table for cable size requirement.

Model	Power Supply Cord	Cable Size	Isolating Devices	Recommended RCD
WH-MXC09J3E5 and WH-MXC12J6E5	1	3 x min 4.0 mm^2	30A	30mA, 2P, type A
	2	3 x min 4.0 mm^2	30A	30mA, 2P, type AC

2. To avoid the cable and cord being damaged by sharp edges, the cable and cord must be passed through the designated holes before being connected to the terminal block.
3. Secure the cable onto the control board with the holder (clamper).

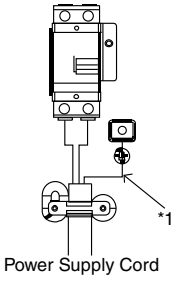


12.1.4.3 Wire Stripping and Connecting Requirement



Terminal screw	Tightening torque cN•m {kgf•cm}
M4	157~196 {16~20}
M5	196~245 {20~25}

*1 - Earth wire must be longer than other cables for safety reasons.



12.1.4.4 Connecting Requirement

For WH-MXC09J3E5

- This equipment's Power Supply 1 complies with IEC/EN 61000-3-12 provided that the short circuit power S_{sc} is greater than or equal to 1900kW at the interface point between the user's supply and the public system. It is the responsibility of the installer or user of the equipment to ensure, by consultation with the distribution network operator if necessary, that the equipment is connected only to a supply with a short circuit power S_{sc} greater than or equal to 1900kW.
- The equipment's Power Supply 1 complies with IEC/EN 61000-3-11 and shall be connected to a suitable supply network, having services current capacity ≥ 100A per phase. Please liaise with supply authority that the service current capacity at the interface point is sufficient for the installation of the equipment.
- The equipment's Power Supply 2 complies with IEC/EN 61000-3-2.
- The equipment's Power Supply 2 complies with IEC/EN 61000-3-3 and can be connected to current supply network.

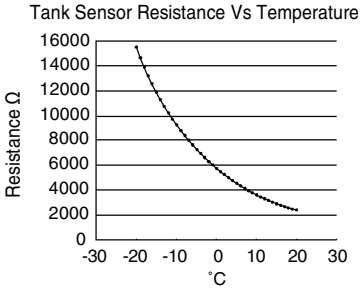
For WH-MXC12J6E5

- This equipment's Power Supply 1 complies with IEC/EN 61000-3-12 provided that the short circuit power S_{sc} is greater than or equal to 1900kW at the interface point between the user's supply and the public system. It is the responsibility of the installer or user of the equipment to ensure, by consultation with the distribution network operator if necessary, that the equipment is connected only to a supply with a short circuit power S_{sc} greater than or equal to 1900kW.
- The equipment's Power Supply 1 complies with IEC/EN 61000-3-11 and shall be connected to a suitable supply network, having services current capacity ≥ 100A per phase. Please liaise with supply authority that the service current capacity at the interface point is sufficient for the installation of the equipment.
- This equipment's Power Supply 2 complies with IEC/EN 61000-3-12.
- The equipment's Power Supply 2 complies with IEC/EN 61000-3-11 and shall be connected to suitable supply network, with the following maximum permissible system impedance Z_{max} = 0.193 ohm (Ω) at the interface. Please liaise with supply authority to ensure that the Power Supply 2 is connected only to a supply of that impedance or less.

12.1.4.5 Connecting with external device (optional)

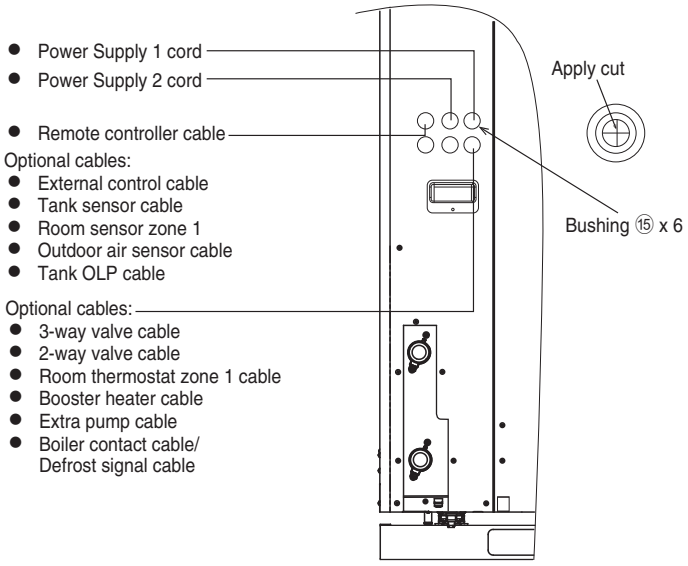
- **All connections shall** follow to the local national wiring standard.
- It is strongly recommended to use manufacturer-recommended parts and accessories for installation.
- For connection to main PCB
  1 Two-way valve shall be spring and electronic type, refer to "Field Supply Accessories" table for details. Valve cable shall be (3 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier, or similarly double insulation sheathed cable.
    * note: - Two-way Valve shall be CE marking compliance component.
    - Maximum load for the valve is 9.8VA.
  2 Three-way valve shall be spring and electronic type. Valve cable shall be (3 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier, or similarly double insulation sheathed cable.
    * note: - Shall be CE marking compliance component.
    - It shall be directed to heating mode when it is OFF.
    - Maximum load for the valve is 9.8VA.
  3 Room thermostat cable must be (4 or 3 x min 0.5 mm^2), of type designation 60245 IEC 57 or heavier cord, or similarly double insulation sheathed cable.
  4 Maximum output power of booster heater shall be ≤ 3 kW. Booster heater cable must be (3 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
  5 Extra pump cable shall be (2 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
  6 Boiler contact cable shall be (2 x min 0.5 mm^2), of type designation 60245 IEC 57 or heavier.
  7 External control shall be connected to 1-pole switch with min 3.0 mm contact gap. Its cable must be (2 x min 0.5 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
    * note: - Switch used shall be CE compliance component.
    - Maximum operating current shall be less than $3A_{rms}$.

8 Tank sensor shall be resistance type, please refer to graph below for the characteristic and details of sensor. Its cable shall be (2 x min 0.3 mm^2), double insulation layer (with insulation strength of min 30V) of PVC-sheathed or rubber-sheathed cable.

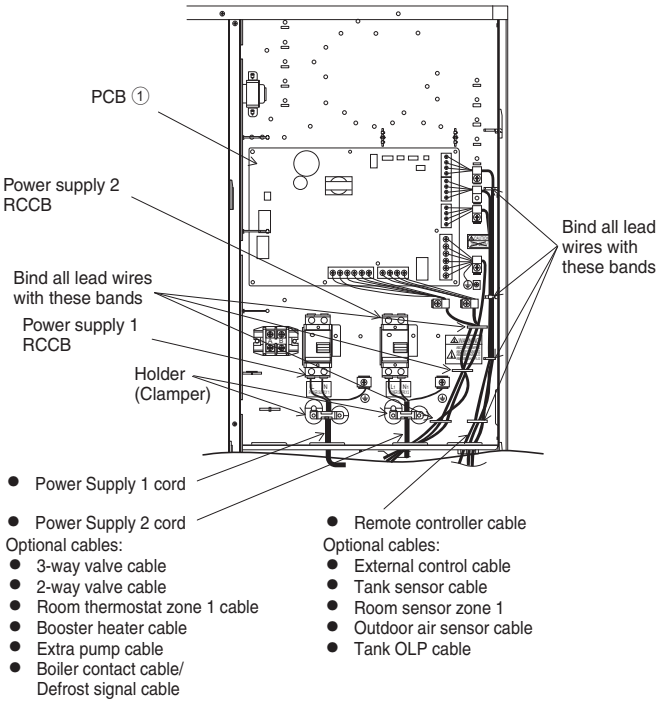


Tank sensor characteristic

9 Room sensor zone 1 cable shall be (2 x min 0.3 mm^2) double insulation layer of PVC-sheathed or rubber-sheathed.
10 Outdoor air sensor cable shall be (2 x min 0.3 mm^2) double insulation layer of PVC-sheathed or rubber-sheathed.
11 Tank OLP cable must be (2 x min 0.5 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.

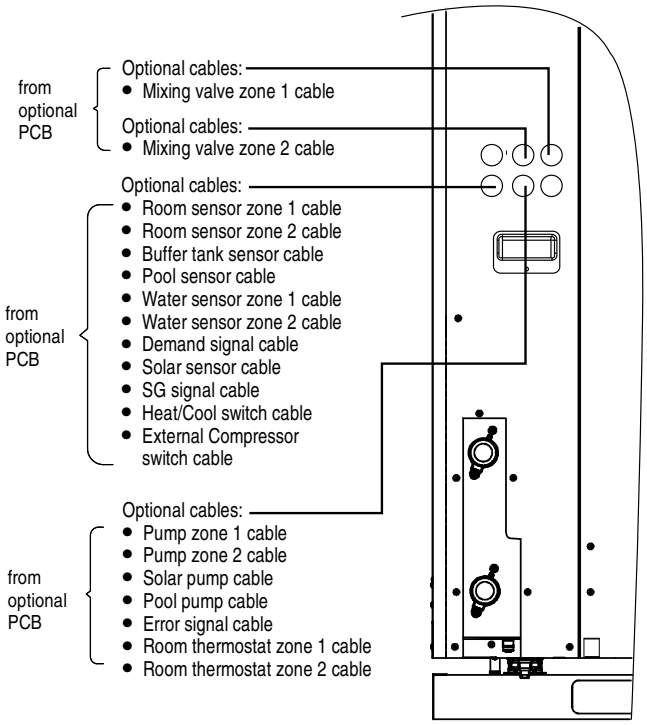


How to guide optional cables and power supply cords to Bushing ⑮

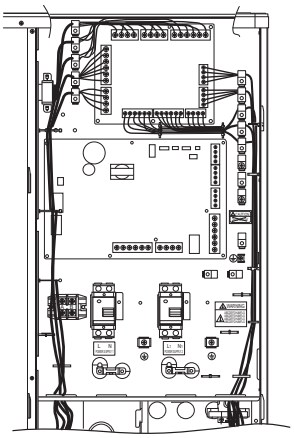


How to guide optional cables and power supply cords
(view without internal wiring)

- For connection to optional PCB
  1 By connecting optional PCB, 2 Zone temperature control can be achieved. Please connect mixing valves, water pumps and thermistors in zone 1 and zone 2 to each terminals in optional PCB. Temperature of each zone can be controlled independently by remote controller.
  2 Pump zone 1 and zone 2 cable shall be (2 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
  3 Solar pump cable shall be (2 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
  4 Pool pump cable shall be (2 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
  5 Room thermostat zone 1 and zone 2 cable shall be (4 x min 0.5 mm^2), of type designation 60245 IEC 57 or heavier.
  6 Mixing valve zone 1 and zone 2 cable shall be (3 x min 1.5 mm^2), of type designation 60245 IEC 57 or heavier.
  7 Room sensor zone 1 and zone 2 cable shall be (2 x min 0.3 mm^2), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
  8 Buffer tank sensor, pool water sensor and solar sensor cable shall be (2 x min 0.3 mm^2), double insulation layer (with insulation strength of minimum 30V) of PVC-sheathed or rubber-sheathed cable.
  9 Water sensor zone 1 and zone 2 cable shall be (2 x min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  10 Demand signal cable shall be (2 x min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  11 SG signal cable shall be (3 x min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  12 Heat/Cool switch cable shall be (2 x min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.
  13 External compressor switch cable shall be (2 x min 0.3 mm^2), double insulation layer of PVC-sheathed or rubber-sheathed cable.



How to guide optional cables to bushing ⑮



- Pump zone 1 cable
- Pump zone 2 cable
- Solar pump cable
- Pool pump cable
- Error signal cable
- Room thermostat zone 1 cable
- Room thermostat zone 2 cable
- Mixing valve zone 1 cable
- Mixing valve zone 2 cable
- Room sensor zone 1 cable
- Room sensor zone 2 cable
- Buffer tank sensor cable
- Pool sensor cable
- Water sensor zone 1 cable
- Water sensor zone 2 cable
- Demand signal cable
- Solar sensor cable
- SG signal cable
- Heat/Cool switch cable
- External Compressor switch cable

How to guide optional cables (view without internal wiring)

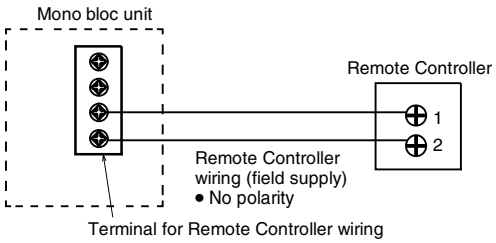
Terminal screw on PCB	Maximum tightening torque cN•m {kgf•cm}
M3	50 {5.1}
M4	120 {12.24}

12.1.5 Installation of Remote Controller

12.1.5.1 Installation Location

- Install at the height of 1 to 1.5 m from the floor (Location where average room temperature can be detected).
- Install vertically against the wall.
- Avoid the following locations for installation.
  1 By the window, etc. exposed to direct sunlight or direct air.
  2 In the shadow or backside of objects deviated from the room airflow.
  3 Location where condensation occurs (The Remote Controller is not moisture proof or drip proof.)
  4 Location near heat source.
  5 Uneven surface.
- Keep distance of 1 m or more from the TV, radio and PC. (Cause of fuzzy image or noise)

12.1.5.2 Remote Controller Wiring

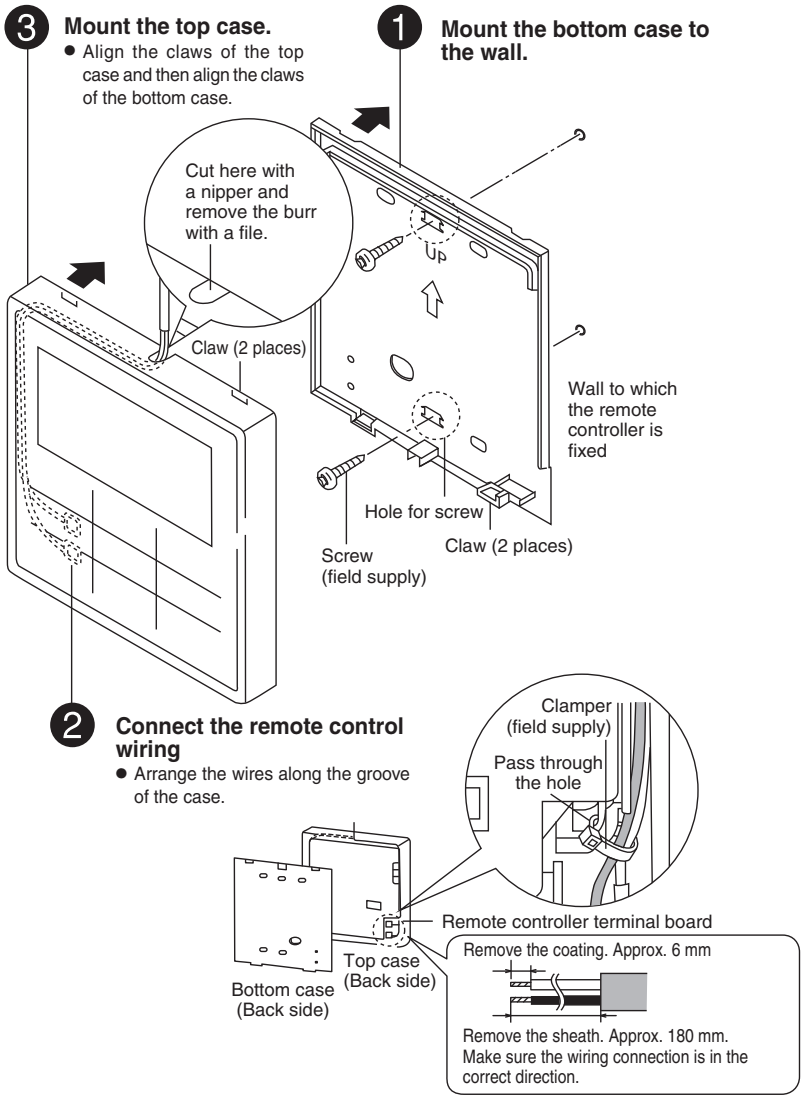


- Remote Controller cable shall be (2 x min 0.3 mm^2), of double insulation PVC-sheathed or rubber sheathed cable. Total cable length shall be 50 m or less.
- Be careful not to connect cables to other terminals (e.g. power source wiring terminal). Malfunction may occur.
- Do not bundle together with the power source wiring or store in the same metal tube. Operation error may occur.

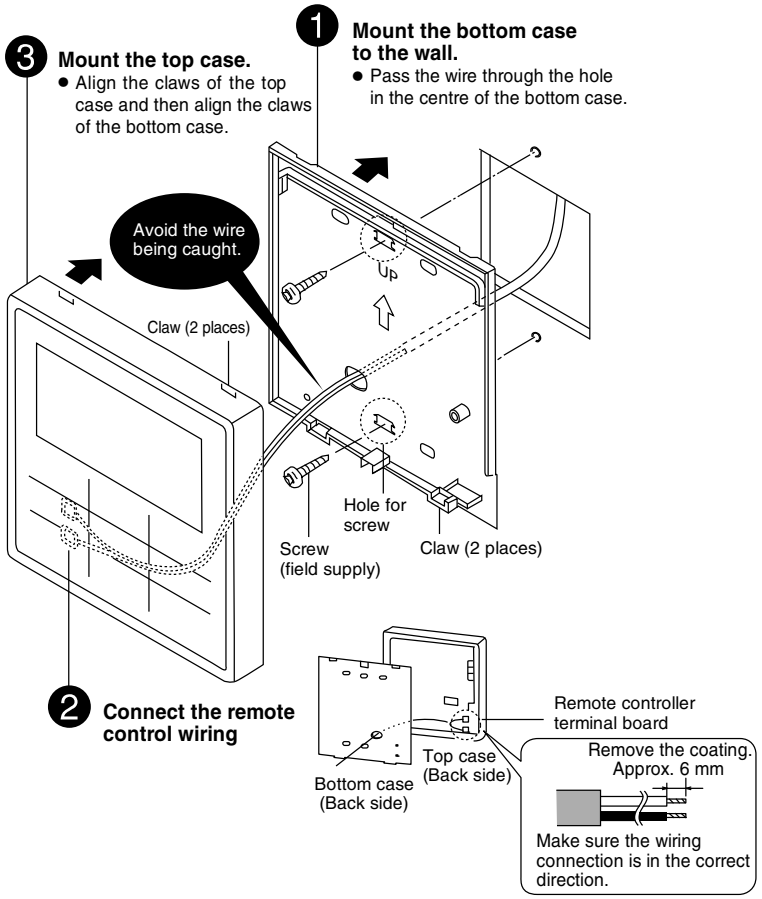
12.1.5.3 Mounting The Remote Controller

For exposed type

Preparation: Make 2 holes for screws using a driver.



For embedded type
Preparation: Make 2 holes for screws using a driver.



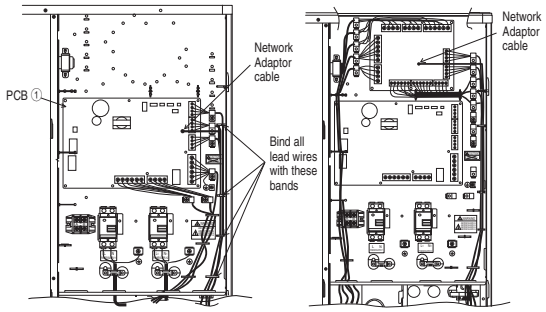
⚠ WARNING

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Work behind the Cabinet Front Plate secured by screws must only be carried out under supervision of qualified contractor, installation engineer or service person.

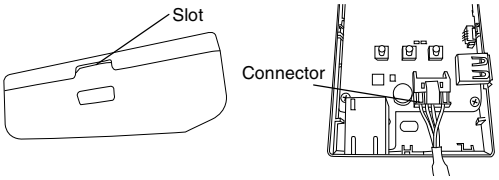
12.1.6 Installation of Network Adaptor and Base Pan Heater

12.1.6.1 Network Adaptor and Installation (Optional)

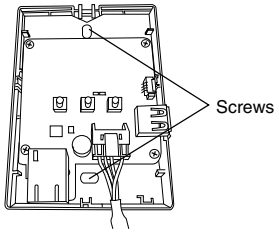
1. **Open the Cabinet front plate, then connect the Network Adaptor Cable to the CN-CNT connector on the printed circuit board.**
   - Pull the cable out of the Mono bloc unit so that there is no pinching.
   - If an optional PCB has been install in the Mono bloc unit, connect the CN-CNT connector to Optional PCB.



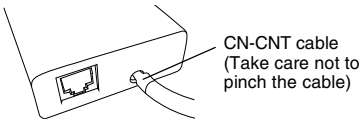
2. **Insert a flat head screwdriver into the slot on the top of the adaptor and remove the cover. Connect the other end of the CN-CNT cable connector to the connector inside the adaptor.**



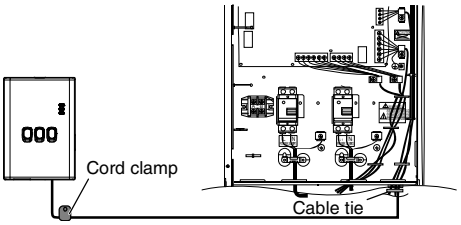
3. **On the wall near the Mono bloc unit, attach the adaptor by screwing screws through the holes in the back cover.**



4. **Pull the CN-CNT cable through the hole in the bottom of the adaptor and re-attach the front cover to the back cover.**



5. **Use the included cord clamp to fix the CN-CNT cable to the wall.**
   Pull the cable around as shown in the diagram so that external forces cannot act on the connector in the adaptor. Use the included cable tie to fix the cables together.

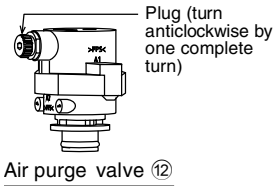


12.1.6.2 Base Pan Heater (Optional)

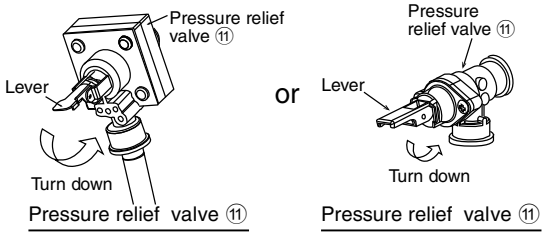
- It is strongly recommended to install a Base Pan Heater (optional) if the Mono bloc unit is installed in cold climate area. Refer the Base Pan Heater (optional) installation instruction for details of installation.

12.1.7 Charging the Water

- Make sure all the piping installations are properly done before carry out below steps.
  1 Open Cabinet front plate to access to the Pressure Relief Valve and Air Purge Valve.
  2 Turn the plug on the Air Purge Valve outlet anticlockwise by one complete turn from fully closed position.



3 Set the Pressure Relief Valve level “DOWN”.



4 Start filling water (with pressure more than 0.1 MPa (1 bar)) to the Mono bloc unit via water inlet. Stop filling water if the free water flow through Pressure Relief Valve drain hose.
5 Turn ON the power supply and make sure Water Pump is running.
6 Check and make sure no water leaking at the tube connecting points.
7 Reinstall the Cabinet front plate by tightening the 2 mounting screws.

12.1.8 Reconfirmation

⚠ WARNING
Be sure to switch off all power supply before performing each of the below checkings. Before obtaining access to terminals, all supply circuits must be disconnected.

*12.1.8.1 Check Water Pressure *(0.1 MPa = 1 bar)*

Water pressure should not lower than 0.05 MPa (with inspects the Water Pressure Gauge). If necessary add tap water into the water circuit.

12.1.8.2 Check Pressure Relief Valve

- Check for correct operation of Pressure Relief Valve by turning on the lever to become horizontal.
- If you do not hear a clacking sound (due to water drainage), contact your local authorized dealer.
- Push down the lever after finish checking.
- In case the water keeps drained out from the unit, switch off the system, and then contact your local authorized dealer.

12.1.8.3 Expansion Vessel Pre Pressure Checking

[Upper limit water volume of the system]
The Mono bloc unit has a build-in Expansion Vessel with 10 L air capacity and initial pressure of 1 bar.

- Without antifreeze agent condition

Total amount of water in the system should be below 200 L.
If the total amount of water is more than 200 L, please add expansion vessel (field supply).

- With antifreeze agent condition

In the case of using antifreeze agent, expansion rate ε is different depending on its maker.
Please refer to the antifreeze agent maker for the expansion rate ε before calculate the upper limit water volume of the system.
The expansion vessel capacity required for the system can be calculated from the formula below.

$$V = \frac{\varepsilon \times Vo}{1 - \frac{98 + P_1}{98 + P_2}}$$

V : Required gas volume <expansion vessel volume L>
Vo : System total water volume <L>
ε : Expansion rate 5→60˚C = (depends on antifreeze agent used)
P_1 : Expansion tank filling pressure = (100) kPa
P_2 : System maximum pressure = 300 kPa
- ( ) Please confirm at actual place
- The gas volume of the sealed type expansion vessel is presented by <V
❍ It's advised to add 10% margin for required gas volume of calculation.

[Adjustment of the initial pressure of the expansion vessel when there is a difference in installation height]
If the height difference between the Mono bloc unit and the highest point of the system water circuit (H) is more than 7m, please adjust the initial pressure of the expansion vessel (Pg) according to the following formula.

Pg= (H*10+30) kPa

12.1.8.4 Check RCCB

Ensure the RCCB set to "ON" condition before check RCCB.
Turn on the power supply to the Mono bloc unit.
This testing could only be done when power is supplied to the Mono bloc unit.

⚠ WARNING
Be careful not to touch parts other than RCCB test button when the power is supplied to Mono bloc unit. Else, electrical shock may happen.

- Push the "TEST" button on the RCCB. The lever would turn down and indicate "0", if it functions normal.
- Contact authorized dealer if the RCCB malfunction.
- Turn off the power supply to the Mono bloc unit.
- If RCCB functions normal, set the lever to "ON" again after testing finish.

This product contains fluorinated greenhouse gasses.
Refrigerant type : R410A (GWP=675)
For WH-MXC09J3E5 & WH-MXC12J6E5
Amount : 1.6kg (1.080 ton CO_2 equivalent)

12.1.9 Test Run

1. Before test run, make sure below items have been checked:-
   - Pipework are properly done.
   - Electric cable connecting work are properly done.
   - Mono bloc unit is filled up with water and trapped air is released.
2. Antifreeze agent must be added into water circuit to prevent freezing of water when outdoor ambient temperatures is low. Recommended antifreeze: Propylene glycol: 40% (equivalent to -20°C)
3. Set ON to the Mono bloc unit and RCCB. Then, for remote control operation please refers to (Mono bloc) Air-to-Water Heatpump's operation instruction.

   Note:
   - During winter turn on the power supply and standby the unit for at least 15 minutes before test run. Allow sufficient time to warm up refrigerant and prevent wrong error code judgement.
4. For normal operation, Water Pressure Gauge reading should be in between 0.05 MPa and 0.3 MPa.
5. After test run, please clean the Magnetic Water Filter Set. Reinstall it after finish cleaning.

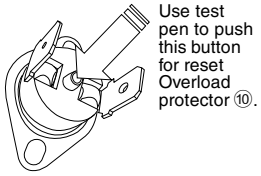
12.1.9.1 Check Water Flow of Water Circuit

Confirm the maximum water flow during main pump operation not less than 15 l/min.
*Water flow can be check through service setup (Pump Max Speed)
[Heating operation at low water temperature with lower water flow may trigger “H75” during defrost process.]

12.1.9.2 Reset Overload Protector

Overload Protector serves the safety purpose to prevent the water over heating. When the Overload Protector trip at high water temperature, take below steps to reset it.

1. Take out the cover.
2. Use a test pen to push the centre button gently in order to reset the Overload Protector.
3. Fix the cover to the original fixing condition.



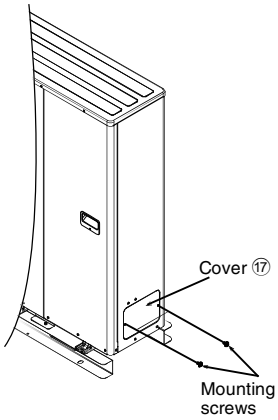
12.1.10 Maintenance

- In order to ensure optimal performance of the unit, seasonal inspections on the unit, functional check of RCCB, field wiring and piping have to be carried out at regular intervals. This maintenance should be carried out by authorized dealer.

12.1.10.1 Maintenance for Magnetic Water Filter Set

1. Remove the Cover by loosening the mounting screws to access to the Magnetic Water Filter Set.
2. Turn OFF power supply.
3. Set the two valves for the Magnetic Water Filter Set to “CLOSE”.
4. Take off the clip, then gently pull out the mesh. Beware of small amount water drain out from it.
5. Clean the mesh with warm water to remove all the stain. Use soft brush if necessary.
6. Reinstall the mesh to the Magnetic Water Filter Set and set back the clip on it.
7. Set the two valves for the Magnetic Water Filter Set to “OPEN”.
8. Turn ON power supply.
9. After cleaning, reinstall the Cover by tightening the mounting screws properly.

⚠ WARNING
Do not add or replace other than R32 type. It may cause product damage, burst, injury and etc. Use compatible R32 tools for refrigerant piping work and refrigerant charging during installation or servicing.



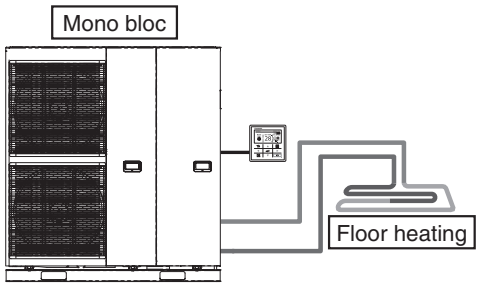
12.2 Variation of system

This section introduces variation of various systems using Air-To-Water Heatpump and actual setting method.

12.2.1 Introduce application related to temperature setting.

12.2.1.1 Temperature setting variation for heating

1. Remote Controller



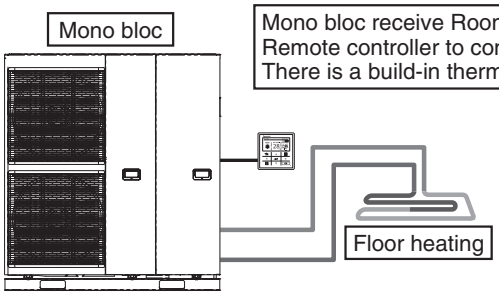
Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Water temperature

Connect floor heating or radiator directly to the Mono bloc.
This is the basic form of the most simple system.

2. Room Thermostat



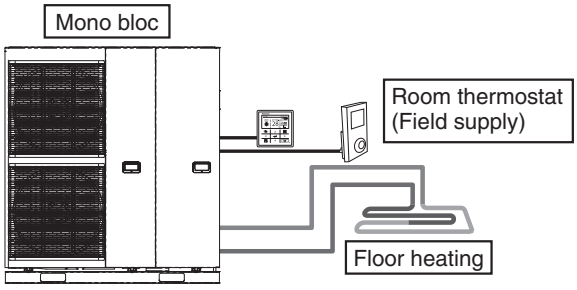
Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Room thermostat
Internal

Connect floor heating or radiator directly to the Mono bloc.
Install the remote controller in the room where floor heating is installed.
This is an application that uses remote controller as Room Thermostat.

3. External Room Thermostat



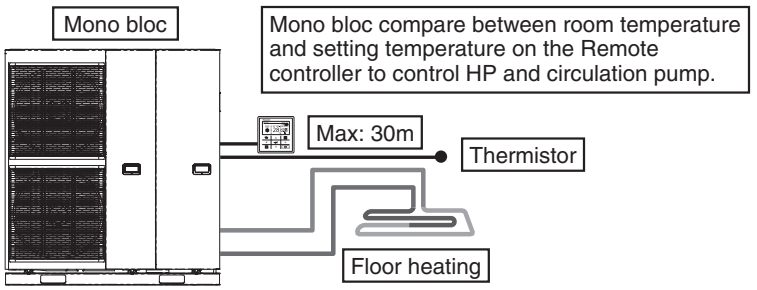
Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

Zone & Sensor:
Room thermostat
(External)

Connect floor heating or radiator directly to Mono bloc.
Install separate external Room Thermostat (field supply) in the room where floor heating is installed.
This is an application that uses external Room Thermostat.

4. Room Thermistor



Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

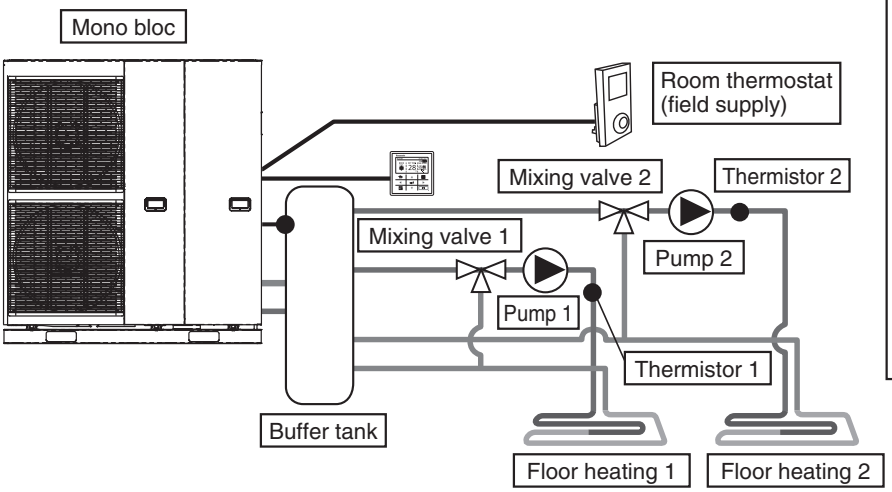
Zone & Sensor:
Room thermistor

Connect floor heating or radiator directly to Mono bloc.
Install separate external room thermistor (specified by Panasonic) in the room where floor heating is installed.
This is an application that uses external room thermistor.

There are 2 kinds of circulation water temperature setting method.
Direct: set direct circulation water temperature (fixed value)
Compensation curve: set circulation water temperature depends on outdoor ambient temperature
In case of Room thermo or Room thermistor, compensation curve can be set.
In this case, compensation curve is shifted according to the thermo ON/OFF situation.
- (Example) If room temperature increasing speed is;
  very slow → shift up the compensation curve
  very fast → shift down the compensation curve

12.2.1.2 Examples of installations

Floor heating 1 + Floor heating 2



Setting of remote controller

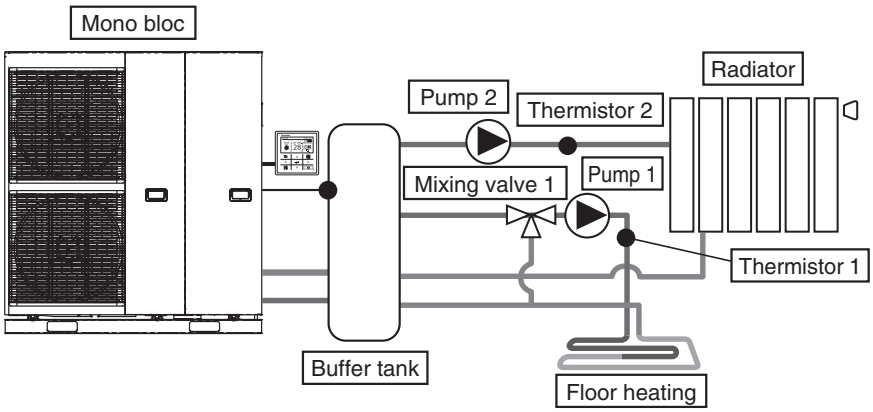
Installer setting
System setup
Optional PCB connectivity - Yes

Zone and Sensor - 2 Zone system
Zone 1:Sensor
Room thermostat
Internal

Zone 2:Sensor
Room
Room thermostat
(External)

Connect floor heating to 2 circuits through buffer tank as shown in the figure.
Install mixing valves, pumps and thermistors (specified by Panasonic) on both circuits.
Install the remote controller in one of the circuit and use it as Room Thermostat.
Install external Room Thermostat (field supply) in another circuit.
Both circuits can set circulation water temperature independently.
Install buffer tank thermistor on buffer tank.
It requires connection setting of buffer
tank and ΔT temperature setting at heating operation separately.
This system requires optional PCB (CZ-NS4P).

Floor heating + Radiator



Setting of remote controller

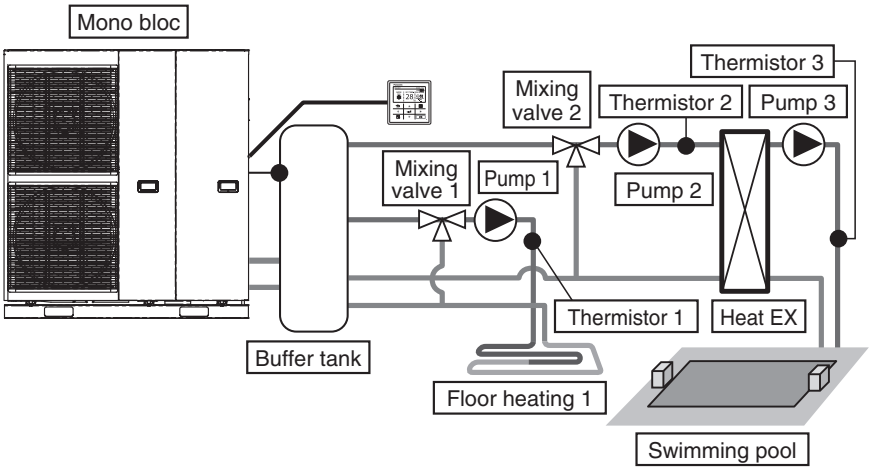
Installer setting
System setup
Optional PCB connectivity - Yes

Zone and Sensor - 2 Zone system
Zone 1:Sensor
Water temperature

Zone 2:Sensor
Room
Water temperature

Connect floor heating or radiator to 2 circuits through buffer tank as shown in figure.
Install pumps and thermistors (specified by Panasonic) on both circuits.
Install mixing valve in the circuit with lower temperature among the 2 circuits.
(Generally, if install floor heating and radiator circuit at 2 zones, install mixing valve in floor heating circuit.)
For temperature setting, select circulation water temperature for both circuits.
Both circuits can set circulation water temperature independently.
Install buffer tank thermistor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately.
This system requires the optional PCB (CZ-NS4P).
Mind that if there is no mixing valve at the secondary side, the circulation water temperature may get higher than setting temperature.

Floor heating + Swimming pool



Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes

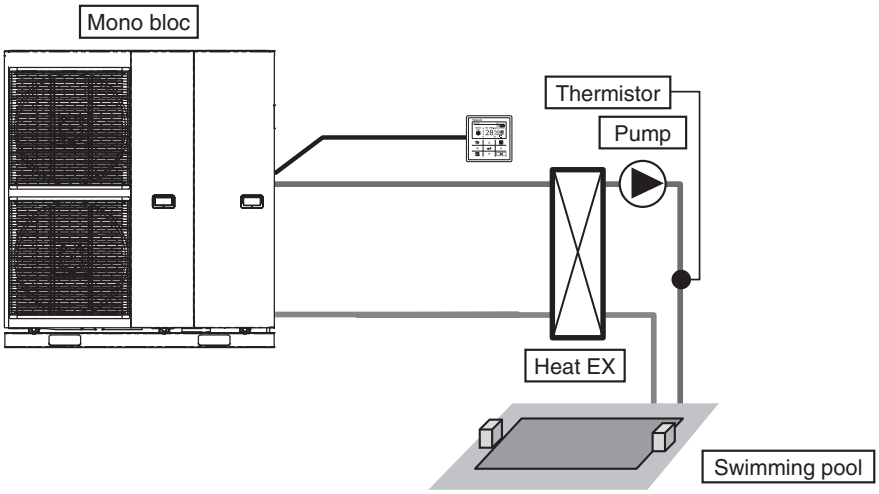
Zone and Sensor - 2 Zone system
Zone 1:Sensor
Room thermostat
Internal

Zone 2
Swimming pool
ΔT

Connect floor heating and swimming pool to 2 circuits through buffer tank as shown in figure.
Install mixing valves, pumps and thermistors (specified by Panasonic) on both circuits.
Then, install additional pool heat exchanger, pool pump and pool sensor on pool circuit.
Install the remote controller in room where floor heating is installed. Circulation water temperature of floor heating and swimming pool can be set independently.
Install buffer tank sensor on buffer tank.
It requires connection setting of buffer tank and ΔT temperature setting at heating operation separately. This system requires the optional PCB (CZ-NS4P).

※ Must connect swimming pool to “Zone 2”.
If it is connected to swimming pool, operation of pool will stop when “Cooling” is operated.

Swimming pool only

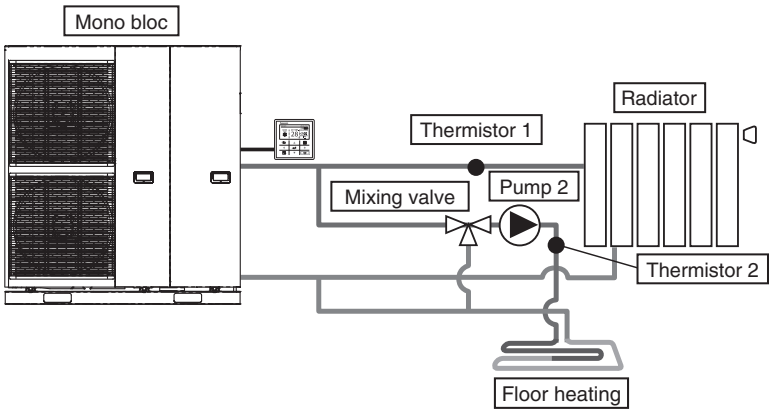


Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Zone and Sensor - 1 Zone system
Zone :Swimming pool
ΔT

This is an application that connects to the swimming pool only.
Connects pool heat exchanger directly to Mono bloc without using buffer tank.
Install pool pump and pool sensor (specified by Panasonic) at secondary side of the pool heat exchanger.
Install the remote controller in room where floor heating is installed.
Temperature of swimming pool can be set independently.
This system requires the optional PCB (CZ-NS4P).
In this application, cooling mode cannot be selected. (not display on remote controller)

Simple 2 zone (Floor heating + Radiator)



Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Zone and Sensor - 2 Zone system
Zone 1:Sensor
Water temperature
Zone 2:Sensor
Room
Water temperature
Operation setup
Heat
ΔT for heating ON – 1°C
Cool
ΔT for cooling ON – 1°C

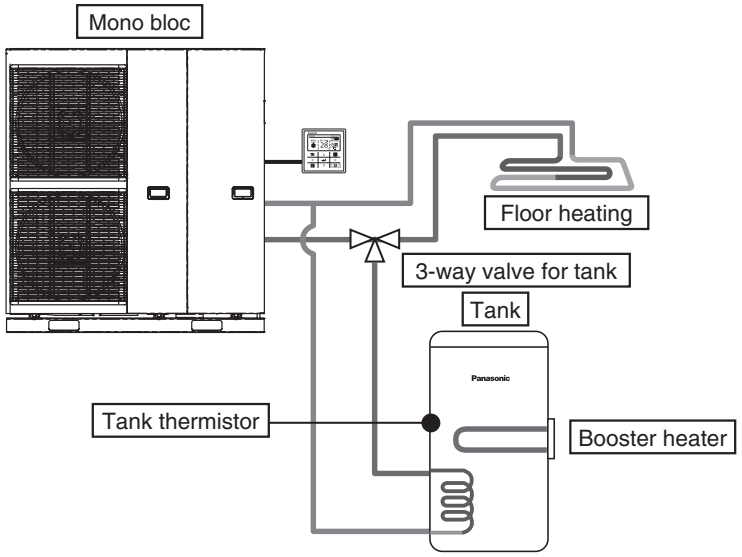
This is an example of simple 2 zone control without using buffer tank.
Built-in pump from Mono bloc served as a pump in zone 1.
Install mixing valve, pump and thermistor (specified by Panasonic) on zone 2 circuit.
Please be sure to assign high temperature side to zone 1 as temperature of zone 1 cannot be adjusted.
Zone 1 thermistor is required to display temperature of zone 1 on remote controller.
Circulation water temperature of both circuits can be set independently.
(However, temperature of high temperature side and low temperature side cannot be reversed)
This system requires the optional PCB (CZ-NS4P).

(NOTE)

- Thermistor 1 does not affect operation directly. But error happens if it is not installed.
  Please adjust flow rate of zone 1 and zone 2 to be in balance. If it is not adjusted correctly, it may affects the performance.
  (If zone 2 pump flow rate is too high, there is possibility that no hot water flowing to zone 1.)
  Flow rate can be confirmed by “Actuator Check” from maintenance menu.

12.2.2 Introduce applications of system that uses optional equipment.

DHW (Domestic Hot Water) Tank connection



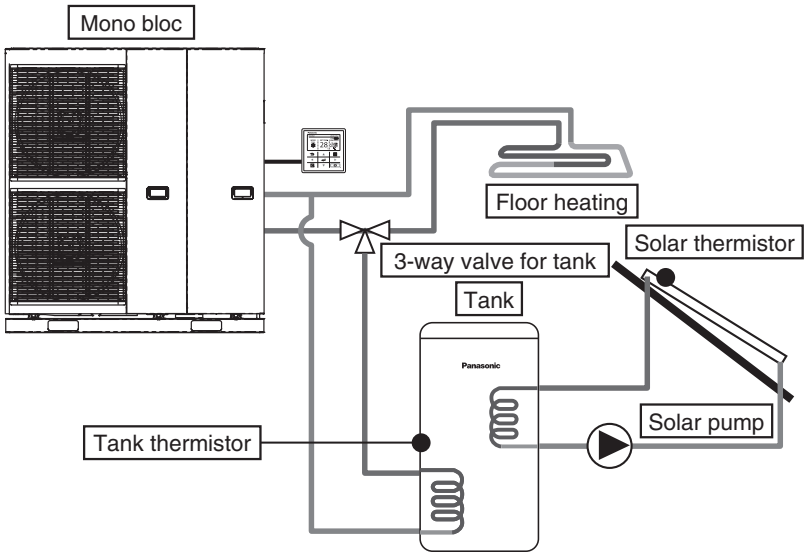
Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - No

Tank connection - Yes

This is an application that connects the DHW tank to the Mono bloc through 3-way valve.
DHW tank's temperature is detected by tank thermistor (specified by Panasonic).

Tank + Solar connection



Setting of remote controller

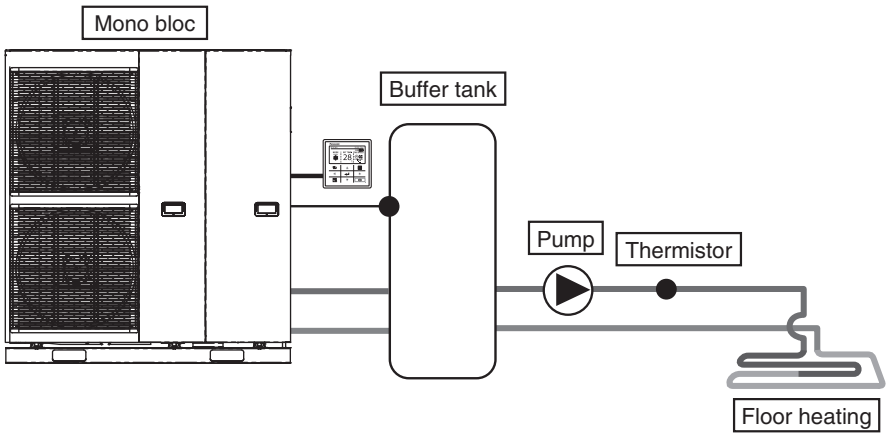
Installer setting
System setup
Optional PCB connectivity - Yes

Tank connection - Yes

Solar connection - Yes
DHW tank
ΔT turn ON
ΔT turn OFF
Antifreeze
Hi limit

This is an application that connects the DHW tank to the Mono bloc through 3-way valve before connect the solar water heater to heat up the tank. DHW tank's temperature is detected by tank thermistor (specified by Panasonic).
Solar panel's temperature is detected by solar thermistor (specified by Panasonic).
DHW tank shall use tank with built-in solar heat exchange coil independently.
Heat accumulation operates automatically by comparing the temperature of tank thermistor and solar thermistor.
During winter season, solar pump for circuit protection will be activated continuously. If does not want to activate the solar pump operation, please use glycol and set the anti-freezing operation start temperature to -20°C.
This system requires optional PCB (CZ-NS4P).

Buffer tank connection

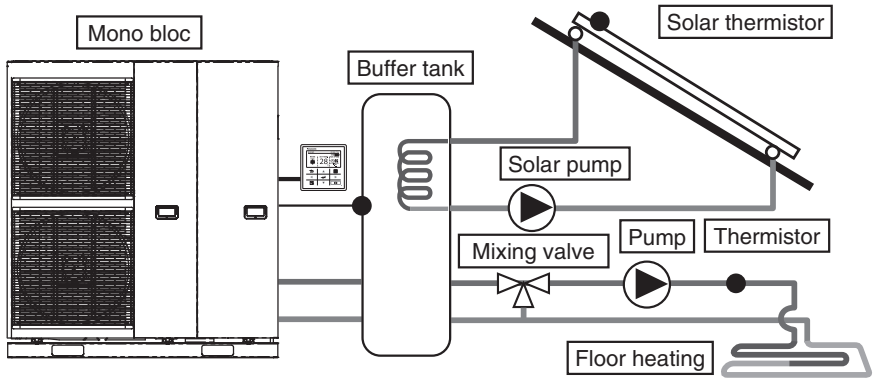


Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Buffer Tank connection - Yes
ΔT for buffer tank

This is an application that connects the buffer tank to the Mono bloc.
Buffer tank's temperature is detected by buffer tank thermistor (specified by Panasonic).
This system requires optional PCB (CZ-NS4P).

Buffer tank + Solar

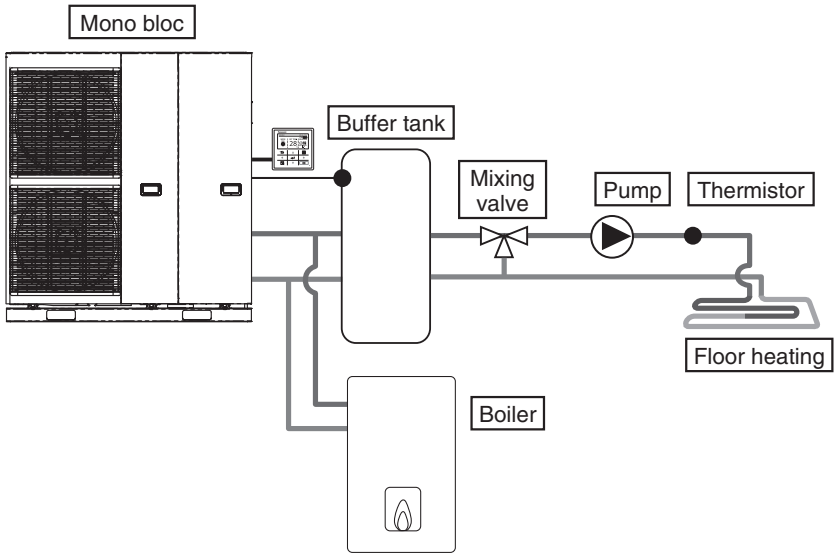


Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Buffer Tank connection - Yes
ΔT for buffer tank
Solar connection - Yes
Buffer tank
ΔT turn ON
ΔT turn OFF
Antifreeze
Hi limit

This is an application that connects the buffer tank to the Mono bloc before connecting to the solar water heater to heat up the tank.
Buffer tank's temperature is detected by buffer tank thermistor (specified by Panasonic).
Solar panel's temperature is detected by solar thermistor (specified by Panasonic).
Buffer tank shall use tank with built-in solar heat exchange coil independently.
During winter season, solar pump for circuit protection will be activated continuously. If does not want to activate the solar pump operation, please use glycol and set the anti-freezing operation start temperature to -20°C.
Heat accumulation operates automatically by comparing the temperature of tank thermistor and solar thermistor.
This system requires optional PCB (CZ-NS4P).

Boiler connection



Setting of remote controller

Installer setting
System setup
Optional PCB connectivity - Yes
Bivalent - Yes
Turn ON: outdoor temp
Control pattern

This is an application that connects the boiler to the Mono bloc, to compensate for insufficient capacity by operate boiler when outdoor temperature drops & heat pump capacity is insufficient.
Boiler is connected parallel with heat pump against heating circuit.
There are 3 modes selectable by remote controller for boiler connection.
Besides that, an application that connects to the DHW tank's circuit to heat up tank's hot water is also possible. (Operation setting of boiler shall be responsible by installer.)
This system requires optional PCB (CZ-NS4P).
Depending on the settings of the boiler, it is recommended to install buffer tank as temperature of circulating water may get higher. (It must connect to buffer tank especially when selecting Advanced Parallel setting.)

⚠ WARNING
Panasonic is NOT responsible for incorrect or unsafe situation of the boiler system.

⚠ CAUTION
Make sure the boiler and its integration in the system complies with applicable legislation. Make sure the return water temperature from the heating circuit to the indoor unit does NOT exceed 55°C. Boiler is turned off by safety control when the water temperature of the heating circuit exceed 85°C.

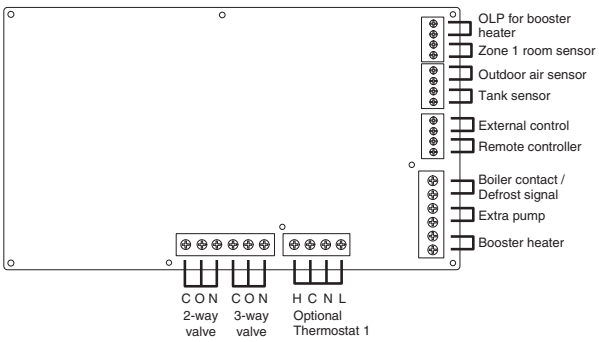
12.3 How to Fix External Device

12.3.1 Connecting Cables Length

When connecting cables between Mono bloc and external devices, the length of the said cables must not exceed the maximum length as shown in the table.

External device	Maximum cables length (m)
Two-way valve	50
Three-way valve	50
Mixing valve	50
Room thermostat	50
Booster heater	50
Extra pump	50
Solar pump	50
Pool pump	50
Pump	50
Boiler contact / Defrost signal	50
External control	50
Tank sensor	30
Room sensor	30
Outdoor air sensor	30
Tank OLP	30
Buffer tank sensor	30
Pool water sensor	30
Solar sensor	30
Water sensor	30
Demand signal	50
SG signal	50
Heat/Cool switch	50
External compressor switch	50

12.3.1.1 Connection of the main PCB



■ Signal inputs

Optional Thermostat	L N =AC230V, Heat, Cool=Thermostat heat, Cool terminal ❊It does not function when using the optional PCB
OLP for booster heater	Dry contact Vcc-Bit1, Vcc-Bit2 open/short (System setup necessary) It is connected to the safety device (OLP) of DHW tank.
External control	Dry contact Open=not operate, Short=operate (System setup necessary) Able to turn ON/OFF the operation by external switch
Remote controller	Connected (Please use 2 cores wire for relocation and extension. Total cable length shall be 50m or less.)

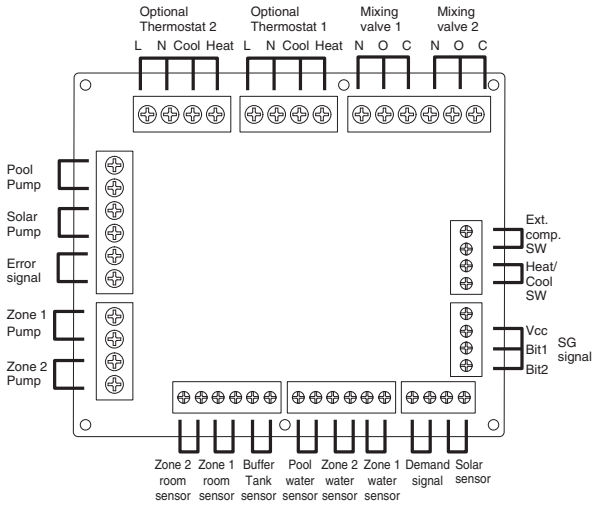
■ Outputs

3-way valve	AC230V N=Neutral Open, Close=direction (For circuit switching when connected to DHW tank)
2-way valve	AC230V N=Neutral Open, Close (Prevent water circuit pass through during cooling mode)
Extra pump	AC230V (Used when Mono bloc pump capacity is insufficient)
Booster heater	AC230V (Used when using booster heater in DHW tank)
Boiler contact / Defrost signal	Dry contact (System setup necessary)

■ Thermistor inputs

Zone 1 room sensor	PAW-A2W-TSRT ❊It does not work when using the optional PCB
Outdoor air sensor	AW-A2W-TSOD (Total cable length shall be 30m or less)
Tank sensor	Please use Panasonic specified part

12.3.1.2 Connection of Optional PCB (CZ-NS4P)



■ Signal inputs

Optional Thermostat	L N =AC230V, Heat, Cool=Thermostat heat, Cool terminal
SG signal	Dry contact Vcc-Bit1, Vcc-Bit2 open/short (System setup necessary) Switching SW (Please connect to the 2 contacts controller)
Heat/Cool SW	Dry contact Open=Heat, Short=Cool (System setup necessary)
External comp.SW	Dry contact Open=Comp.ON, Short=Comp.OFF (System setup necessary)
Demand signal	DC 0~10V (System setup necessary) Please connect to the DC 0~10V controller.

■ Outputs

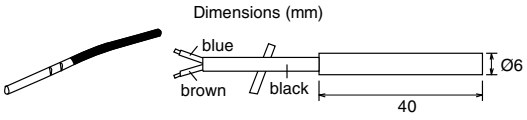
Mixing valve	AC230V N=Neutral Open, Close=mixture direction Operating time: 30s~120s
Pool pump	AC230V
Solar pump	AC230V
Zone pump	AC230V

■ Thermistor inputs

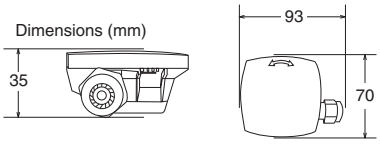
Zone room sensor	PAW-A2W-TSRT
Buffer tank sensor	PAW-A2W-TSBU
Pool water sensor	PAW-A2W-TSHC
Zone water sensor	PAW-A2W-TSHC
Solar sensor	PAW-A2W-TSSO

12.3.1.3 Recommended External Device Specification

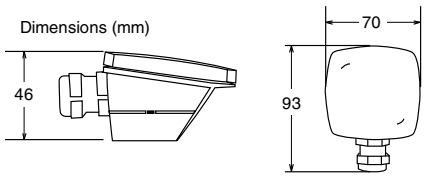
- This section explains about the external devices (optional) recommended by Panasonic. Please always ensure to use the correct external device during system installation.
- For optional sensor.
  1 Buffer tank sensor: PAW-A2W-TSBU
  Use for measurement of the buffer tank temperature.
  Insert the sensor into the sensor pocket and paste it on the buffer tank surface.



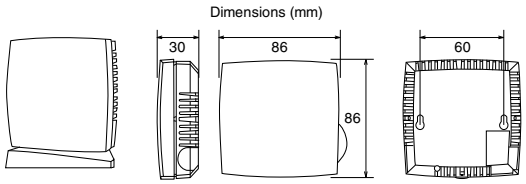
  2 Zone water sensor: PAW-A2W-TSHC
  Use to detect the water temperature of the control zone.
  Mount it on the water piping by using the stainless steel metal strap and contact paste (both are included).



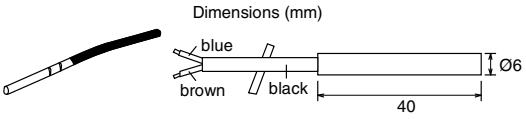
  3 Outdoor sensor: PAW-A2W-TSOD
  If the installation location of the outdoor unit is exposed to direct sunlight, the outdoor air temperature sensor will be unable to measure the actual outdoor ambient temperature correctly.
  In this case, optional outdoor temperature sensor can be fixed at a suitable location to more accurately measure ambient temperature.



  4 Room sensor: PAW-A2W-TSRT
  Install the room temperature sensor to the room which requires room temperature control.



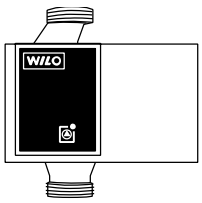
  5 Solar sensor: PAW-A2W-TSSO
  Use for measurement of the solar panel temperature.
  Insert the sensor into the sensor pocket and paste it on the solar panel surface.



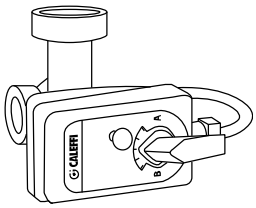
  6 Please refer to the table below for sensor characteristic of the sensors mentioned above.

Temperature (°C)	Resistance (kΩ)
150	0.147
140	0.186
130	0.236
120	0.302
110	0.390
100	0.511
90	0.686
80	0.932
70	1.279
65	1.504
60	1.777
55	2.106
50	2.508
45	3.003
40	3.615
35	4.375
30	5.326
25	6.523
20	8.044
15	9.980
10	12.443
5	15.604
0	19.70
-5	25.05
-10	32.10
-15	41.45
-20	53.92
-25	70.53
-30	93.05
-35	124.24
-40	167.82

- For optional pump.
  Power supply: AC230V/50Hz, <500W
  Recommended part: Yonos 25/6: made by Wilo

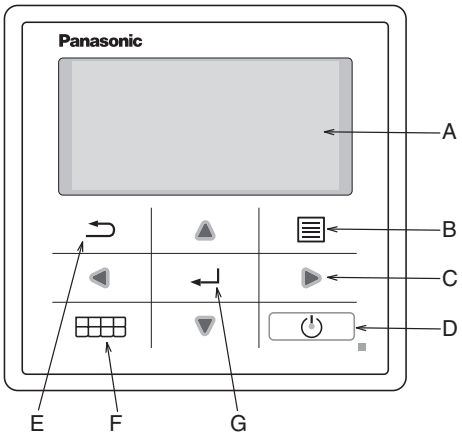


- For optional mixing valve.
  Power supply: AC230V/50Hz (input open/output close)
  Operating time: 30s~120s
  Recommended part: 167032: made by Caleffi

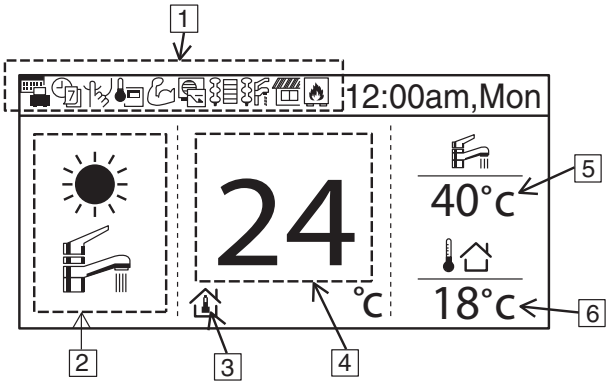


12.4 System Installation

12.4.1 Remote Controller Outline



	Name	Function
A:	Main screen	Display information
B:	Menu	Open/Close main menu
C:	Triangle (Move)	Select or change item
D:	Operate	Start/Stop operation
E:	Back	Back to previous item
F:	Quick Menu	Open/Close Quick menu
G:	OK	Confirm



	Name	Function
1:	Function icon	Display set function/status

Holiday mode
Weekly timer
Quiet mode
Remote controller room thermostat
Powerful mode
Demand control
Room heater
Tank heater
Solar
Boiler

2: Mode — Display set mode/current status of mode

Heating
Cooling
Auto
Hot water supply
Auto heating
Auto cooling
Heat pump operating

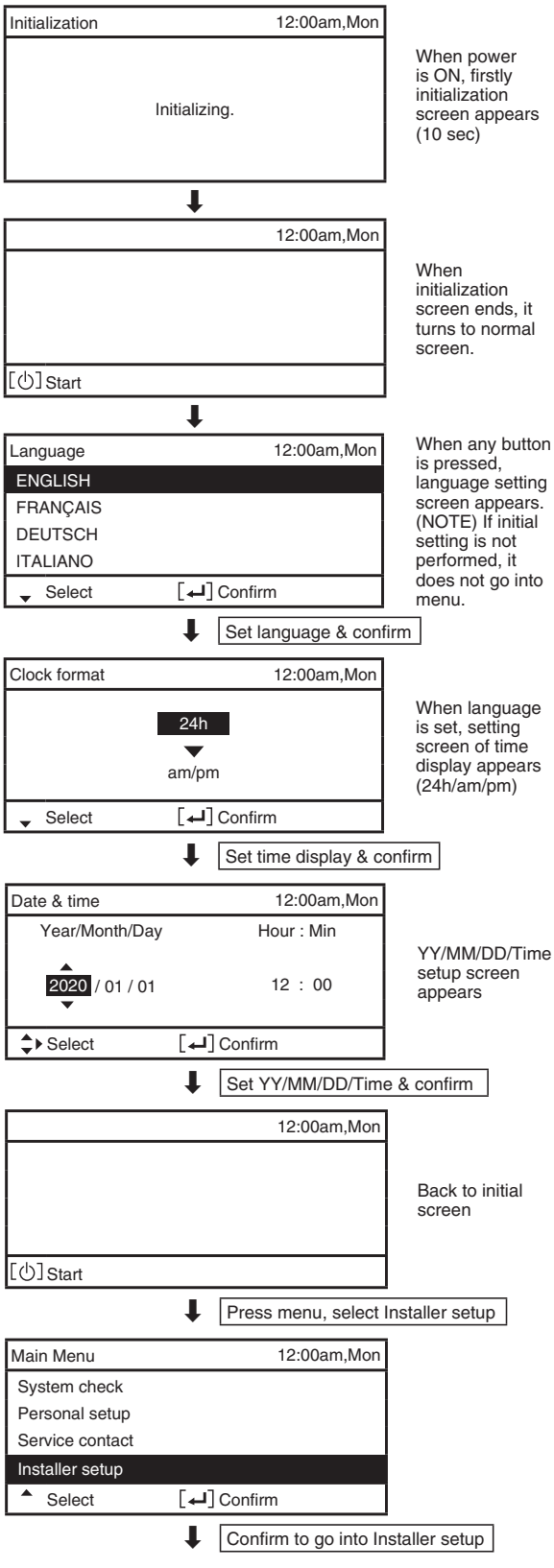
3: Temp setting
Set room temp
Compensation curve
Set direct water temp
Set pool temp

4: Display Heat temp — Display current heating temperature (it is set temperature when enclosed by line)

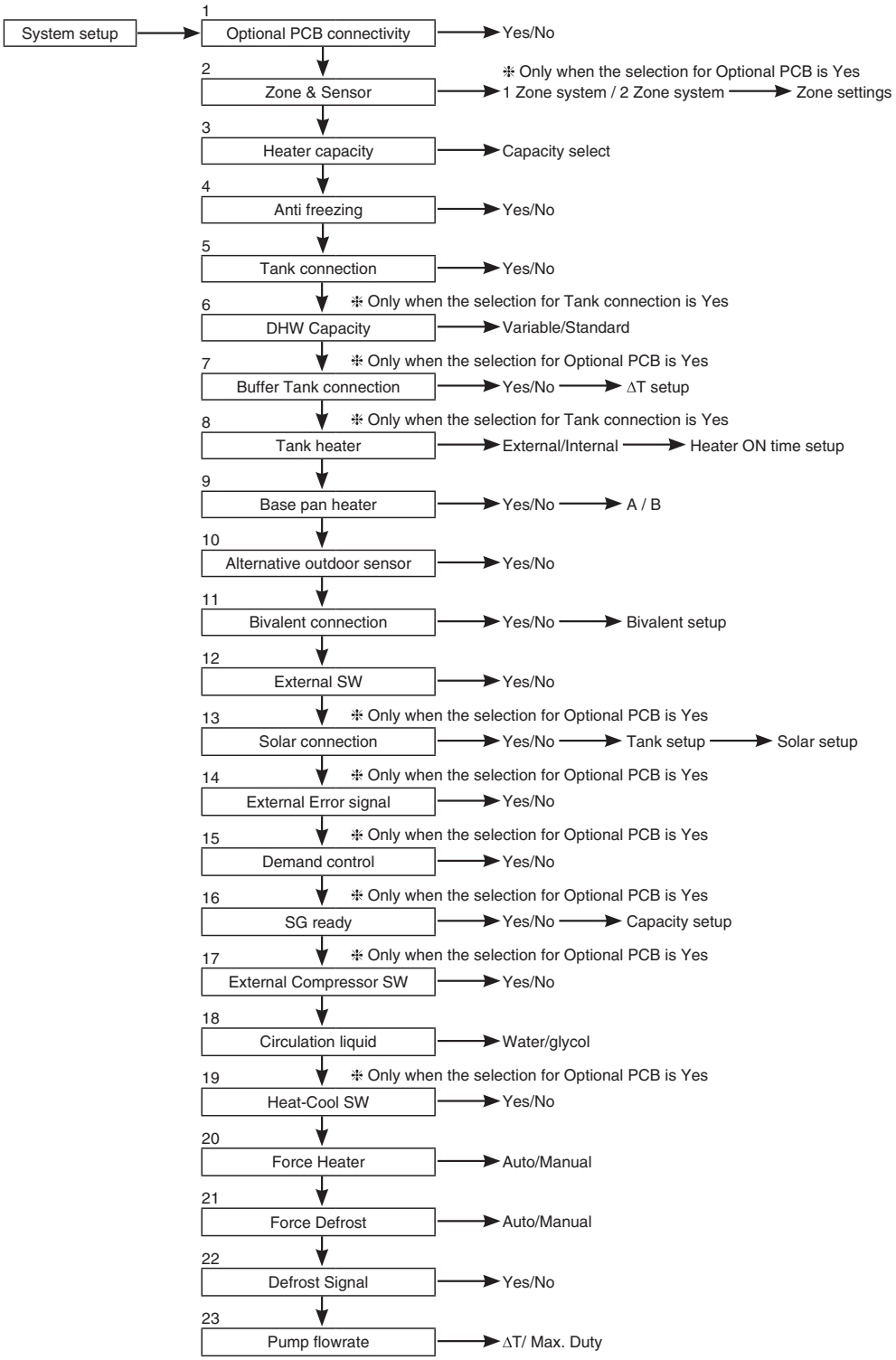
5: Display tank temp — Display current tank temperature (it is set temperature when enclosed by line)

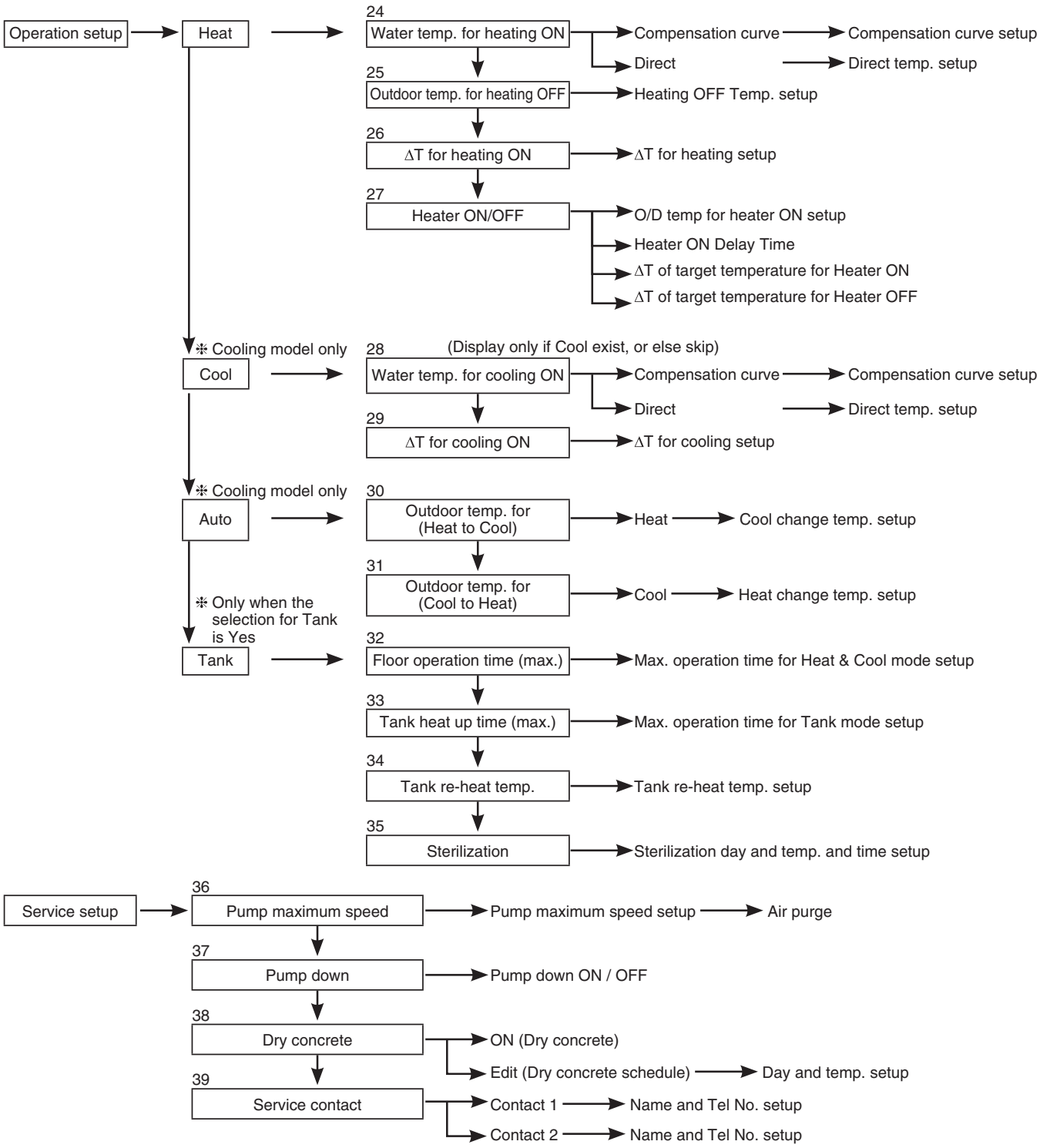
6: Outdoor temp — Display outdoor temp

First time of power ON (Start of installation)



12.4.2 Installer Setup



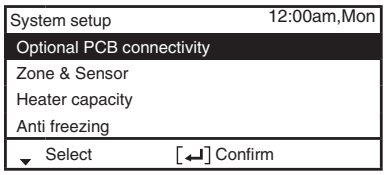

Operation setup
Heat
24
Water temp. for heating ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
25
Outdoor temp. for heating OFF
Heating OFF Temp. setup
26
ΔT for heating ON
ΔT for heating setup
27
Heater ON/OFF
O/D temp for heater ON setup
Heater ON Delay Time
ΔT of target temperature for Heater ON
ΔT of target temperature for Heater OFF
❊ Cooling model only
Cool
28
(Display only if Cool exist, or else skip)
Water temp. for cooling ON
Compensation curve
Compensation curve setup
Direct
Direct temp. setup
29
ΔT for cooling ON
ΔT for cooling setup
❊ Cooling model only
Auto
30
Outdoor temp. for (Heat to Cool)
Heat
Cool change temp. setup
31
Outdoor temp. for (Cool to Heat)
Cool
Heat change temp. setup
❊ Only when the selection for Tank is Yes
Tank
32
Floor operation time (max.)
Max. operation time for Heat & Cool mode setup
33
Tank heat up time (max.)
Max. operation time for Tank mode setup
34
Tank re-heat temp.
Tank re-heat temp. setup
35
Sterilization
Sterilization day and temp. and time setup
Service setup
36
Pump maximum speed
Pump maximum speed setup
Air purge
37
Pump down
Pump down ON / OFF
38
Dry concrete
ON (Dry concrete)
Edit (Dry concrete schedule)
Day and temp. setup
39
Service contact
Contact 1
Name and Tel No. setup
Contact 2
Name and Tel No. setup

12.4.3 System Setup

1. Optional PCB connectivity | Initial setting: No

If function below is necessary, please purchase and install optional PCB.
Please select Yes after installing optional PCB.

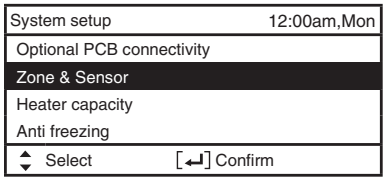
- 2-zone control
- Pool
- Buffer tank
- Solar
- External error signal output
- Demand control
- SG ready
- Stop heat source unit by external SW



2. Zone & Sensor | Initial setting: Room and Water temp.

If no Optional PCB connectivity
Select sensor of room temperature control from the following 3 items
① Water temperature (Circulation water temperature)
② Room thermostat (Internal or External)
③ Room thermistor

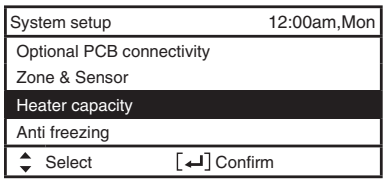
When there is Optional PCB connectivity
① Select either 1 zone control or 2 zone control.
If it is 1 zone, select either room or pool, select sensor
If it is 2 zone, after select sensor of zone 1, select either room or pool for zone 2, select sensor
(NOTE) In 2 zone system, pool function can be set at zone 2 only.



3. Heater capacity | Initial setting: Depend on model

If there is built-in Heater, set the selectable heater capacity.

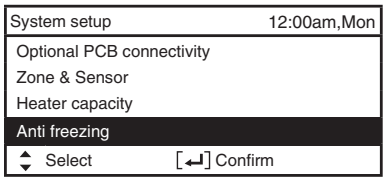
(NOTE) There are models which cannot select heater capacity.



4. Anti freezing | Initial setting: Yes

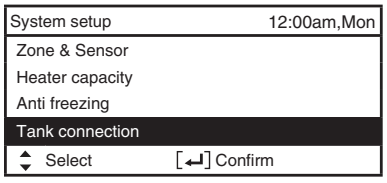
Operate anti-freezing of water circulation circuit.
If select Yes, when the water temperature is reaching its freezing temperature, the circulation pump will start up. If the water temperature does not reach the pump stop temperature, back-up heater will be activated.

(NOTE) If set No, when the water temperature is reaching its freezing temperature or below 0°C, the water circulation circuit may freeze and cause malfunction.



5. Tank connection | Initial setting: No

Select whether it is connected to hot water tank or not.
If set Yes, it becomes setting that uses hot water function.
Hot water temperature of tank can be set from main screen.



6. DHW Capacity | Initial setting: Variable

Variable DHW capacity setting normally run with efficient boiling which is energy saving heating. But while hot water usage high and tank water temperature low, variable DHW mode will run with fast heat up which heat up the tank with high heating capacity.
If standard DHW capacity setting is selected, heat pump run with heating rated capacity at tank heat up operation.

System setup	12:00am,Mon
Heater capacity	
Anti freezing	
Tank connection	
DHW Capacity	
Select	[↵] Confirm

7. Buffer Tank connection | Initial setting: No

Select whether it is connected to buffer tank for heating or not.
If buffer tank is used, please set Yes.
Connect buffer tank thermistor and set, ΔT (ΔT use to increase primary side temp against secondary side target temp).
(NOTE) Does not display if there is no Optional PCB.
If the buffer tank capacity is not so large, please set larger value for ΔT.

System setup	12:00am,Mon
Anti freezing	
Tank connection	
DHW Capacity	
Buffer tank connection	
Select	[↵] Confirm

8. Tank heater | Initial setting: Internal

Select to use either built-in heater or external heater as heater for hot water tank.
If heater is installed on tank, please select External.

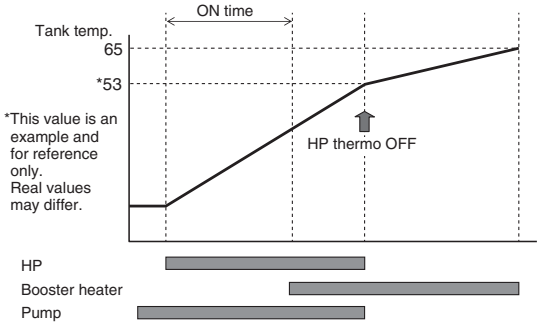
(NOTE) Does not display if there is no tank for hot water supply.

Please set "Tank heater" to "ON" in the "Function setup" from remote controller when using heater to boil the tank.

System setup	12:00am,Mon
Tank connection	
DHW Capacity	
Buffer tank connection	
Tank heater	
Select	[↵] Confirm

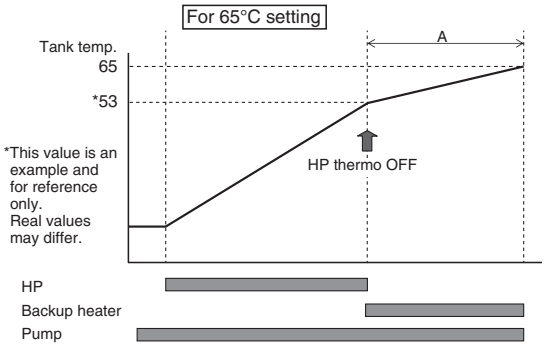
External A setting which is using booster heater installed on DHW tank to boil the tank.
The permissible heater capacity is 3kW and below.
The operation to boil the tank with heater is as below.
In addition, be sure to set suitable "Tank heater: ON time"

For 65°C setting



Internal A setting which is using backup heater of Mono bloc to boil the tank.
The operation to boil the tank with heater is as below.

For 65°C setting



9. Base pan heater | Initial setting: No

Select whether Base pan heater is installed or not.
If set Yes, select to use either heater A or B.

A: Turn on Heater when heating with defrost operation only
B: Turn on Heater at heating

System setup	12:00am,Mon
DHW Capacity	
Buffer tank connection	
Tank heater	
Base pan heater	
Select	[↵] Confirm

10. Alternative outdoor sensor | Initial setting: No

Set Yes if outdoor sensor is installed.
Controlled by optional outdoor sensor without reading the outdoor sensor of heat pump unit.

System setup	12:00am,Mon
Buffer tank connection	
Tank heater	
Base pan heater	
Alternative outdoor sensor	
Select	[↵] Confirm

11. Bivalent connection | Initial setting: No

System setup	12:00am,Mon
Tank heater	
Base pan heater	
Alternative outdoor sensor	
Bivalent connection	
Select	[↵] Confirm

Set if heat pump linked with boiler operation.
Connect the start signal of the boiler in boiler contact terminal (main PCB).
Set Bivalent connection to YES.
After that, please begin setting according to remote controller instruction.
Boiler icon will be displayed on remote controller top screen.

After Bivalent connection Set YES, there is two option of control pattern to be select, (SG Ready / Auto)

1) SG ready (Only available to set when Optional PCB set to YES)
   - SG Ready input from Optional PCB terminal control ON/OFF of boiler and heat pump as below condition

SG signal		Operation pattern
Vcc-bit1	Vcc-bit2	
Open	Open	Heat pump OFF, Boiler OFF
Short	Open	Heat pump ON, Boiler OFF
Open	Short	Heat pump OFF, Boiler ON
Short	Short	Heat pump ON, Boiler ON

* This bivalent SG ready input is sharing same terminal as [16. SG ready] connection. Only one of these two setting can be set at the same time. When one is set, another setting will reset to not set.

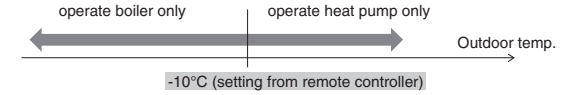
2) Auto (If Optional PCB no Set, bivalent control pattern will set to this auto as default value)

There are 3 different modes in the boiler operation. Movement of each modes are shown below.

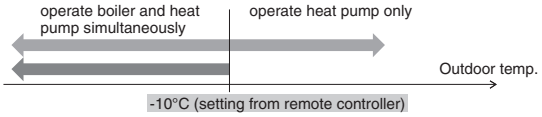
① Alternative (switch to boiler operation when drops below setting temperature)
② Parallel (allow boiler operation when drops below setting temperature)
③ Advanced Parallel (able to slightly delay boiler operation time of parallel operation)

When the boiler operation is “ON”, “boiler contact” is “ON”, “_”(underscore) will be displayed below the boiler icon.
Please set target temperature of boiler to be the same as heat pump temperature.
When boiler temperature is higher than heat pump temperature, zone temperature cannot be achieved if mixing valve is not installed.
This product only allows one signal to control the boiler operation. Operation setting of boiler shall be responsible by installer.

Alternative mode

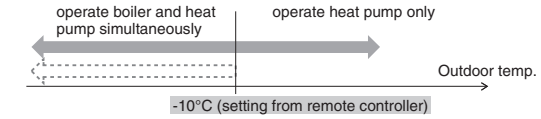


Parallel mode



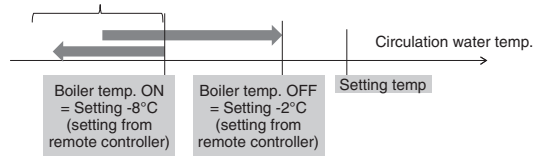
Advanced Parallel mode

For heating

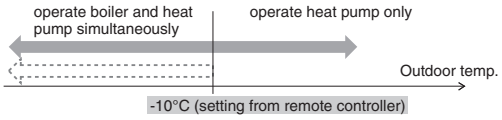


Although heat pump operates but water temperature does not reach this temperature for more than 30 mins (setting from remote controller)

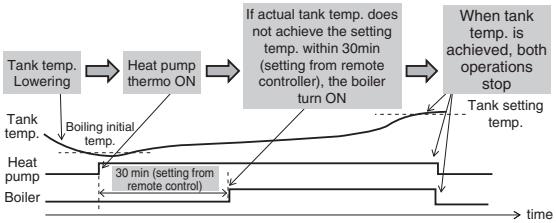
AND



For DHW tank



AND



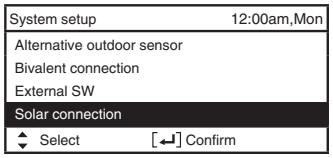
In Advanced Parallel mode, setting for both heating and tank can be made simultaneously. During operation of “Heating/Tank” mode, when each time the mode is switched, the boiler output will be reset to OFF. Please have good understanding on the boiler control characteristic in order to select the optimal setting for the system.

12. External SW | Initial setting: No

System setup	12:00am,Mon
Base pan heater	
Alternative outdoor sensor	
Bivalent connection	
External SW	
Select	[↵] Confirm

Able to turn ON/OFF the operation by external switch.

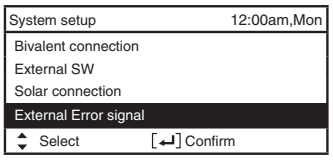
13. Solar connection | Initial setting: No



Set when solar water heater is installed.

Setting include items below.
① Set either buffer tank or DHW tank for connection with solar water heater.
② Set temperature difference between solar panel thermistor and buffer tank or DHW tank thermistor to operate the solar pump.
③ Set temperature difference between solar panel thermistor and buffer tank or DHW tank thermistor to stop the solar pump.
④ Anti-freezing operation start temperature (please change setting based on usage of glycol.)
⑤ Solar pump stop operation when it exceeds high limit temperature (when tank temperature exceed designated temperature (70~90°C))

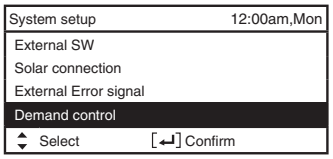
14. External Error signal | Initial setting: No



Set when external error display unit is installed.
Turn on Dry Contact SW when error happened.

(NOTE) Does not display when there is no Optional PCB.
When error occurs, error signal will be ON.
After turn off "close" from the display, error signal will still remain ON.

15. Demand control | Initial setting: No



Set when there is demand control.
Adjust terminal voltage within 1 ~ 10 V to change the operating current limit.

(NOTE) Does not display when there is no Optional PCB.

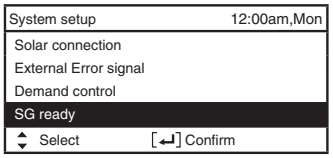
Analog input [V]	Rate [%]
0.0	not activate
0.1 ~ 0.6	not activate
0.7	10
0.8	10
0.9 ~ 1.1	10
1.2	15
1.3	15
1.4 ~ 1.6	15
1.7	20
1.8	20
1.9 ~ 2.1	20
2.2	25
2.3	25
2.4 ~ 2.6	25
2.7	30
2.8	30
2.9 ~ 3.1	30
3.2	35
3.3	35
3.4 ~ 3.6	35
3.7	40
3.8	40

Analog input [V]	Rate [%]
3.9 ~ 4.1	40
4.2	45
4.3	45
4.4 ~ 4.6	45
4.7	50
4.8	50
4.9 ~ 5.1	50
5.2	55
5.3	55
5.4 ~ 5.6	55
5.7	60
5.8	60
5.9 ~ 6.1	60
6.2	65
6.3	65
6.4 ~ 6.6	65
6.7	70
6.8	70
6.9 ~ 7.1	70
7.2	75
7.3	75

Analog input [V]	Rate [%]
7.4 ~ 7.6	75
7.7	80
7.8	80
7.9 ~ 8.1	80
8.2	85
8.3	85
8.4 ~ 8.6	85
8.7	90
8.8	90
8.9 ~ 9.1	90
9.2	95
9.3	95
9.4 ~ 9.6	95
9.7	100
9.8	100
9.9 ~	100

*A minimum operating current is applied on each model for protection purpose.
*0.2 voltage hysteresis is provided.
*The value of voltage after 2nd decimal point are cut off.

16. SG ready | Initial setting: No



Switch operation of heat pump by open-short of 2 terminals.
Setting belows are possible

SG signal		Working pattern
Vcc-bit1	Vcc-bit2	
Open	Open	Normal
Short	Open	Heat pump and Heater OFF
Open	Short	Capacity 1
Short	Short	Capacity 2

Capacity setting 1
- DHW capacity ___%
- Heating capacity ___%
- Cooling capacity ___°C

Capacity setting 2
- DHW capacity ___%
- Heating capacity ___%
- Cooling capacity ___°C

} Set by SG ready setting of remote controller

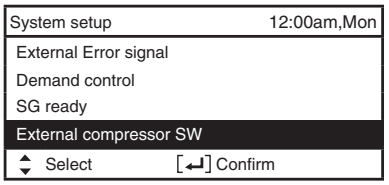
(When SG ready set to YES, Bivalent control pattern will set to Auto.)

17. External Compressor SW | Initial setting: No

Set when external compressor SW is connected.
SW is connected to external devices to control power consumption, Open Signal will stop compressor's operation. (Heating operation etc. are not cancelled).

(NOTE) Does not display if there is no Optional PCB.

If follow Swiss standard power connection, need to turn on DIP SW (SW2 pin3) of main unit PCB. Short/Open signal used to ON/OFF tank heater (for sterilization purpose)

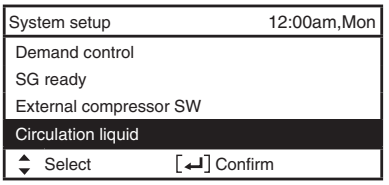


18. Circulation Liquid | Initial setting: Water

Set circulation of heating water.

There are 2 types of settings, water and glycol.

(NOTE) Please set glycol when using anti-freeze liquid.
It may cause error if setting is wrong.

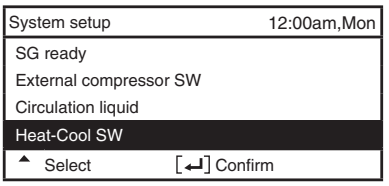


19. Heat-Cool SW | Initial setting: Disable

Able to switch (fix) heating & cooling by external switch.

(Open) : Fix at Heating (Heating +DHW)
(Short) : Fix at Cooling (Cooling +DHW)
(NOTE) This setting is disabled for model without Cooling.
(NOTE) Does not display if there is no Optional PCB.

Timer function cannot be used. Cannot use Auto mode.

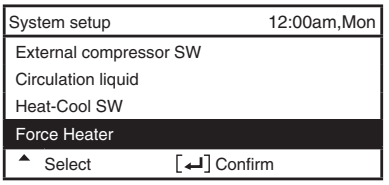


20. Force Heater | Initial setting: Manual

Under manual mode, user can turn on force heater through quick menu.

If selection is 'auto', force heater mode will turn automatically if pop up error happen during operation.
Force heater will operate follow the latest mode selection, mode selection is disable under force heater operation.

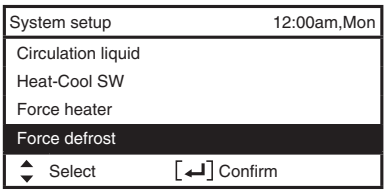
Heater source will ON during force heater mode.



21. Force Defrost | Initial setting: Manual

Under manual code, user can turn on force defrost through quick menu.

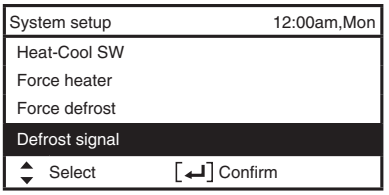
If selection is 'auto', outdoor unit will run defrost operation once if heat pump have long hour of heating without any defrost operation before at low ambient condition.
(Even auto is selected, user still can turn on force defrost through quick menu)



22. Defrost signal | Initial setting: No

Defrost signal sharing same terminal as bivalent contact in main board. When defrost signal set to YES, bivalent connection reset to NO. Only one function can be set between defrost signal and bivalent.

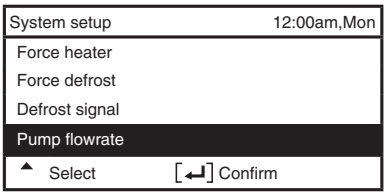
When defrost signal set to YES, during defrost operation is running at outdoor unit defrost signal contact turn ON. Defrost signal contact turn OFF after defrost operation end.
(Purpose of this contact output is to stop indoor fan coil or water pump during defrost operation).



23. Pump flowrate | Initial setting: ΔT

If pump flowrate setting is ΔT, unit adjust pump duty to get different of water inlet and outlet base on setting on * ΔT for heating ON and * ΔT for cooling ON in operation setup menu during room side operation.

If pump flowrate setting is set to Max. duty, unit will set the pump duty to the set duty at *Pump maximum speed in service setup menu during room side operation.



12.4.4 Operation Setup

Heat

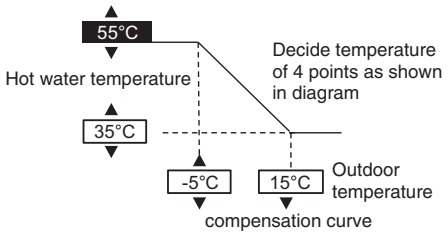
24. Water temp. for heating ON | Initial setting: compensation curve

Set target water temperature to operate heating operation.
Compensation curve: Target water temperature change in conjunction with outdoor ambient temperature change.
Direct: Set direct circulation water temperature.

Regardless of the above setting, the water temperature will be limited as follows.

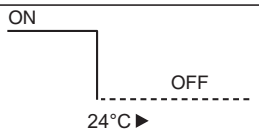
Outdoor temperature	Maximum water temperature limit
-20°C	55°C
-15 ~ 0°C	60°C
5 ~ 20°C	65°C
25 ~ 35°C	60°C

However, only when ΔT is set to 15°C, the water temperature setting of 61°C or higher will take effect.
In 2 zone system, zone 1 and zone 2 water temperature can be set separately.



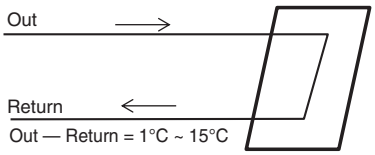
25. Outdoor temp. for heating OFF | Initial setting: 24°C

Set outdoor temp to stop heating.
Setting range is 5°C ~ 35°C



26. ΔT for heating ON | Initial setting: 5°C

Set temp difference between out temp & return temp of circulating water of Heating operation.
When temp gap is enlarged, it is energy saving but less comfort. When the gap gets smaller, energy saving effect gets worse but it is more comfortable.
Setting range is 1°C ~ 15°C



27. Heater ON/OFF

a. Outdoor temp. for heater ON | Initial setting: 0°C

Set outdoor temp when back-up heater starts to operate.
Setting range is -20°C ~ 15°C

User shall set whether to use or not to use heater.

b. Heater ON delay time | Initial setting: 30 minutes

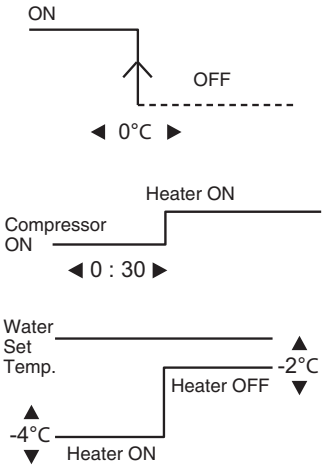
Set delay time from compressor ON for heater to turn ON if not achieve water set temperature.
Setting range is 10 minutes ~ 60 minutes

c. Heater ON: ΔT of target Temp | Initial setting: -4°C

Set water temperature for heater to turn on at heat mode.
Setting range is -10°C ~ -2°C

d. Heater OFF: ΔT of target Temp | Initial setting: -4°C

Set water temperature for heater to turn off at heat mode.
Setting range is -8°C ~ 0°C

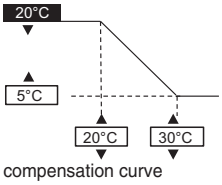


Cool

28. Water temp. for cooling ON | Initial setting: Compensation curve

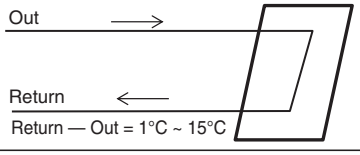
Set target water temperature to operate cooling operation.
Compensation curve: Target water temperature change in conjunction with outdoor ambient temperature change.
Direct: Set direct circulation water temperature.

In 2 zone system, zone 1 and zone 2 water temperature can be set separately.



29. ΔT for cooling ON | Initial setting: 5°C

Set temp difference between out temp & return temp of circulating water of Cooling operation.
When temp gap is enlarged, it is energy saving but less comfort. When the gap gets smaller, energy saving effect gets worse but it is more comfortable.
Setting range is 1°C ~ 15°C

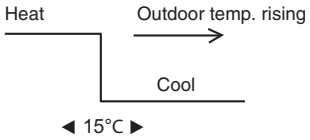


Auto

30. Outdoor temp. for (Heat to Cool) Initial setting: 15°C

Set outdoor temp that switches from heating to cooling by Auto setting.
Setting range is 5°C ~ 25°C

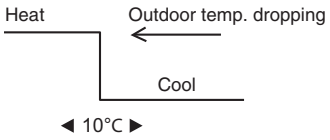
Timing of judgement is every 1 hour



31. Outdoor temp. for (Cool to Heat) Initial setting: 10°C

Set outdoor temp that switches from Cooling to Heating by Auto setting.
Setting range is 5°C ~ 25°C

Timing of judgement is every 1 hour

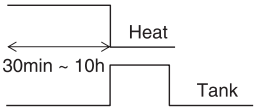


Tank

32. Floor operation time (max.) Initial setting: 8h

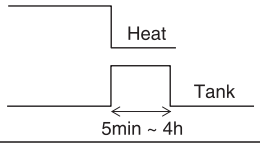
Set max operating hours of heating.
When max operation time is shortened, it can boil the tank more frequently.

It is a function for Heating + Tank operation.



33. Tank heat up time (max.) Initial setting: 60min

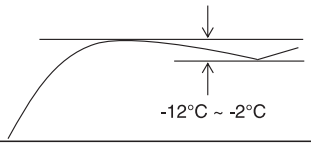
Set max boiling hours of tank.
When max boiling hours are shortened, it immediately returns to Heating operation, but it may not fully boil the tank.



34. Tank re-heat temp. Initial setting: -8°C

Set temp to perform reboil of tank water.

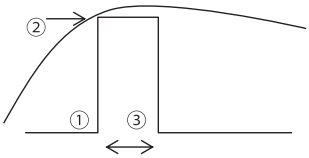
Setting range is -12°C ~ -2°C



35. Sterilization Initial setting: 65°C 10min

Set timer to perform sterilization.
① Set operating day & time. (Weekly timer format)
② Sterilization temp (55~75°C ❄ If use back-up heater, it is 65°C)
③ Operation time (Time to run sterilization when it reached setting temp 5min ~ 60min)

User shall set whether to use or not to use sterilization mode.

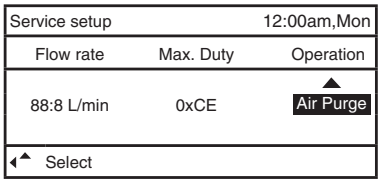


12.4.5 Service Setup

36. Pump maximum speed Initial setting: Depend on model

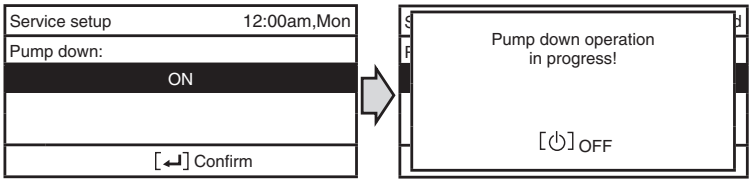
Normally setting is not necessary.
Please adjust when need to reduce the pump sound etc.
Besides that, it has Air Purge function.

When *Pump flow setting is Max. Duty, this duty set is the fix pump duty run during room side operation.



37. Pump down

Operate pump down operation

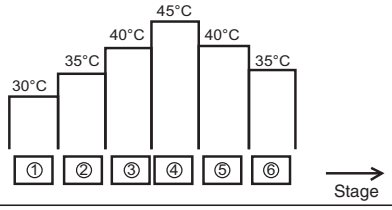


38. Dry concrete

Operate concrete curing operation.
Select Edit, set temp for every stage (1~99 1 is for 1 day).
Setting range is 25~55°C

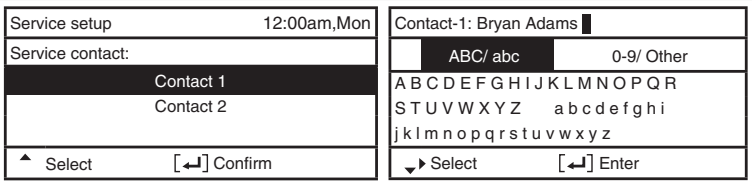
When it is turned ON, dry concrete starts.

When it is 2 zone, it dries both zones.



39. Service contact

Able to set name & tel no. of contact person when there is breakdown etc. or client has trouble. (2 items)



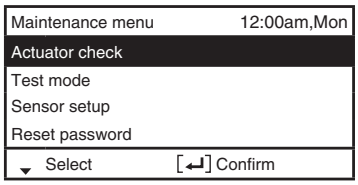
12.5 Service and maintenance

If forget Password and cannot operate remote controller

Press ⮌ + ↵ + ▶ for 5 sec.
Password unlock screen appears, press Confirm and it shall reset.
Password will become 0000. Please reset it again.
(NOTE) Only display when it is locked by password.

Maintenance menu

Setting method of Maintenance menu



Press ⮌ + ↵ + ▶ for 5 sec.

Items that can be set

① Actuator check (Manual ON/OFF all functional parts)
(NOTE) As there is no protection action, please be careful not to cause any error when operating each part (do not turn on pump when there is no water etc.)

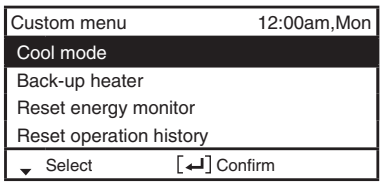
② Test mode (Test run)
Normally it is not used.

③ Sensor setup (offset gap of detected temp of each sensor within -2~2°C range)
(NOTE) Please use only when sensor is deviated.
It affects temperature control.

④ Reset password (Reset password)

Custom menu

Setting method of Custom menu



Please press ☰ + ▼ + ◀ for 10 sec.

Items that can be set

① Cool mode (Set With/Without Cooling function) Default is without
(NOTE) As with/without Cool mode may affect electricity application, please be careful and do not simply change it.
In Cool mode, please be careful if piping is not insulated properly, dew may form on pipe and water may drip on the floor and damage the floor.

② Backup heater (Use/Do not use Backup heater)
(NOTE) It is different from to use/not to use backup heater set by client. When this setting is used, heater power on due to protection against frost will be disabled. (Please use this setting when it is required by utility company.)
By using this setting, it cannot defrost due to low Heating's setting temp and operation may stop (H75)
Please set under the responsibility of installer.
When it stops frequently, it may be due to insufficient circulation flow rate, setting temp of heating is too low etc.

③ Reset energy monitor (delete memory of Energy monitor)
Please use when moving house and handover the unit.

④ Reset operation history (delete memory of operation history).
Please use when moving house and handover the unit.